Music and the Piano From the Ground Up

Notes, Scales, Chords, Rhythm and Reading Music, Explained for Complete Beginners

A First-Principles Reader

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Preface

Who this book is for

This book is for adults who have never had a music lesson. You do not need to know what a note, a chord or a beat is. We start from the very first key and explain every word the first time it appears.

By the end you will know your way around a piano keyboard. You will be able to build scales and chords, keep a steady beat, read simple sheet music, and play your first short pieces with both hands.

How to use this book

Music is something you hear, so this book is meant to be heard. In the web edition, every example has a small round play button next to it. Tap it and you hear exactly what the text describes. Try it now with the note called middle C.

The web edition also has keyboards you can tap. When the text says "press this key", you can press it on the screen and hear it.

A real instrument helps but is not required. A cheap electronic keyboard or a free piano app on your phone is enough. If you have nothing, read with the play buttons and point at the pictures.

Go slowly. Each chapter is short and builds on the one before. When a box says "Try it at the keyboard", give it a minute before reading on.

A two-minute experiment

Here is a taste of what is coming. Do not worry about the names yet; Chapter 1 explains them.

1. Find a pair of black keys near the middle of the keyboard. The white key directly to the left of the pair is called C. Put your right thumb on it.
2. Count four keys to the right, counting every key, black and white, but not the one you start on. Put your middle finger there. That key is E.
3. Count three more keys to the right. Put your little finger there. That key is G.
4. Press all three keys together. This is a C major chord. It sounds bright and settled.
5. Keep your thumb and little finger where they are. Move only your middle finger one key to the

left, onto a black key. Press the three keys again. This is a C minor chord. It sounds darker and more thoughtful.

Hear the two one after the other. One finger moved by one key, and the whole mood changed. That is the spirit of this book: a few clear patterns, seen on the keys and heard with your ears, instead of long lists to memorize.

Where we are going

- *Chapter 1, The keyboard.* Find every note on the piano, name it, and measure the distance between notes.
- *Chapter 2, Scales and keys.* Build the major scale from one simple pattern, and see what "the key of G" means.
- *Chapter 3, Chords.* Build major and minor chords by counting keys, and find the chords that belong together.
- *Chapter 4, Smooth chord changes.* Rearrange chords so your hand barely moves, and accompany a song with two hands.
- *Chapter 5, Rhythm.* Count time: beats, note lengths, rests and time signatures.
- *Chapter 6, Reading notes.* Read notes on the staff, the five lines of sheet music.
- *Chapter 7, Your first pieces.* Put it all together, first a melody, then both hands, with a simple practice routine.

Why music is in a mathematics curriculum

Music is full of patterns: repeating groups of keys, distances you can count, cycles that come back to where they started. You do not need any mathematics to read this book. Counting up to twelve is enough.

If you do enjoy mathematics, look for the boxes titled "For the mathematically curious". They explain the numbers behind the sound, such as why some notes blend and others clash. They are optional, and in the web edition they start closed. Skip them freely.

Part I

The Keyboard, Scales and Keys

Chapter 1

The Keyboard: Finding Every Note

A piano can look overwhelming: 88 keys, black and white, with no labels. The good news is that it is built from one small pattern repeated over and over. Learn that pattern once and you can find any note on any piano.

This chapter teaches you to find and name every key, to tell the same note in different places, and to measure the distance between two notes by counting keys.

1.1 Notes: low and high

Press any key and you hear a **note**: one musical sound. How high or low a note sounds is called its **pitch**. A bird sings at a high pitch; a big drum booms at a low pitch.

On a piano, pitch is laid out from left to right. Keys on the left sound low. Keys on the right sound high. Each key sounds a little higher than the key directly to its left.

Hear a low note, a middle note and a high note. Now hear eight neighboring white keys climbing upward, like walking up a staircase.

Try it at the keyboard

Play a key far to the left, then one far to the right. Then walk up a few white keys from the middle, one at a time, and listen to each note get a little higher.

1.2 The pattern of black keys

Look at the black keys. They are not spread out evenly. They come in groups: a group of two black keys, then a gap, then a group of three, then a gap, then two again, then three, all the way along the keyboard.

This pattern of two and three is your map. Every group of two looks the same, and every group of three looks the same. That is how pianists find their way without labels or stickers.

One group of two, one group of three, and the white keys around them make a block of twelve keys:

seven white and five black. The whole piano repeats this block of twelve keys, from the low end to the high end.

1.3 Finding C and F

Two white keys are easy to find from the black-key groups:

- The white key directly to the left of a group of *two* black keys is called **C**.
- The white key directly to the left of a group of *three* black keys is called **F**.

There is a C to the left of every group of two, so the piano has many Cs. They all sound alike, only higher or lower. Hear three of them. In the same way there is an F before every group of three.

Try it at the keyboard

Find every C on your keyboard, from left to right, by spotting each group of two black keys. Play each one. Then do the same for every F, using the groups of three.

1.4 The seven letter names

Once you have C, the other white keys follow in alphabetical order. Moving to the right from C, the white keys are named C, D, E, F, G, A and B. Then the next white key is C again, and the letters repeat.

Music uses only these seven letters, A to G. After G, the names go back to A. There is no H.

Hear the white keys from C up to the next C, with their names: C, D, E, F, G, A, B and C.

Figure 1.1 shows how the names sit around the black keys. Each group of black keys gives you a few landmarks:

- D sits between the two black keys of a group of two. E is directly to the right of that group.
- G and A sit between the black keys of a group of three. B is directly to the right of that group.

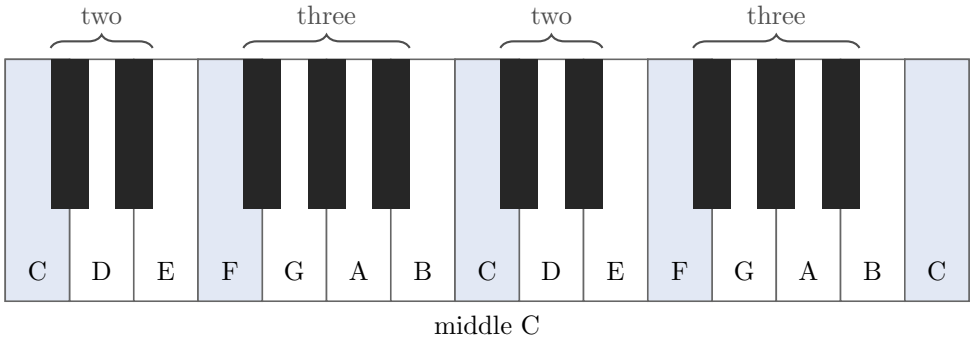


Figure 1.1: Two groups of twelve keys. Every C sits directly left of a group of two black keys, and every F directly left of a group of three. The C in the middle is middle C.

Try it at the keyboard

Put a finger on any D: it is the white key inside a group of two. Now find any B: directly right of a group of three. Play the white keys from D up to the next D and say each name out loud.

1.5 Octaves and octave numbers

Play a C, then the next C to the right. The second one is higher, yet the two sound strangely alike, as if they were the same note in a higher voice. This is why they share a name.

The distance from one note to the next note with the same name is called an **octave**. Going up an octave means going up twelve keys, black and white together. Hear C up an octave, and then both together; they blend so well they almost sound like one note.

To tell the many Cs apart, each one gets a number, called its **octave number**. The C nearest the middle of the piano is called **middle C**, and it is written C4. The next C up is C5, and the next C down is C3. Hear C3, C4 and C5.

The number belongs to all the notes from a C up to the next B. So the white keys directly above middle C are D4, E4, F4 and so on up to B4, and then C5 starts a new octave. The B directly below middle C is B3.

Finger numbers

Piano music numbers your fingers, so it can tell you which finger to use. On both hands the thumb is finger 1, the index finger is 2, the middle finger is 3, the ring finger is 4, and the little finger is 5.

Try it at the keyboard

Put your right thumb on middle C. Rest fingers 2, 3, 4 and 5 on the next white keys: D, E, F and G. Play them one at a time, fingers 1–2–3–4–5, then back down. This place for your hand is called C position.

1.6 Half steps and whole steps

Now we can measure distances. The smallest step on the piano is from one key to the very next key, with nothing in between. This is called a **half step**.

The next key might be black or white. From C to the black key directly right of it is a half step. From that black key to D is another half step. Hear two half steps going up from C.

Two half steps make a **whole step**. A whole step skips exactly one key. From C to D is a whole step, because it skips the black key between them.

The rule is to count every key, black and white. Moving to the next key, whatever its color, is one half step. Figure 1.2 shows both kinds of step.

Look closely at E and F in Figure 1.2. There is no black key between them. So E to F is already a

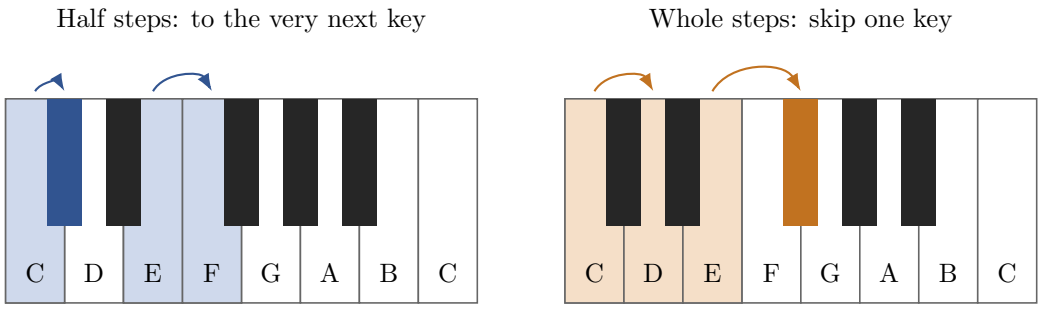


Figure 1.2: Left: C to the black key beside it, and E to F, are both half steps. Right: C to D, and E to the black key directly right of F, are both whole steps.

half step, even though both keys are white. The same is true of B and C. Hear both: E to F, and B to C.

Common trap

It is tempting to think that any two neighboring white keys are a whole step apart. Most are, but E–F and B–C are only half steps, because no black key sits between them. A whole step up from E skips F and lands on the black key directly right of F. A whole step up from B lands on the black key directly right of C.

Try it at the keyboard

Start on C and climb by half steps, touching every key, black and white, up to the next C. You should play 13 keys and take 12 half steps. Then climb from C by whole steps, skipping one key each time.

1.7 Sharps and flats: names for the black keys

The black keys are named after their white neighbors, using two signs:

- A **sharp** (written ♯) means one half step higher, one key to the right. C♯ (said "C sharp") is the key directly right of C.
- A **flat** (written ♭) means one half step lower, one key to the left. D♭ (said "D flat") is the key directly left of D.

The black key between C and D is directly right of C and directly left of D. So it has two names: C♯ and D♭. Both names mean the same key and the same sound. Figure 1.3 shows the two names of every black key.

Which name do musicians use? It depends on the music around it, and Chapter 2 gives the rule. For now, both are correct.

A sharp or flat does not always land on a black key. E♯ means one key right of E, which is the white key F. C♭ means one key left of C, which is the white key B. These names are rare, but you will meet them later.

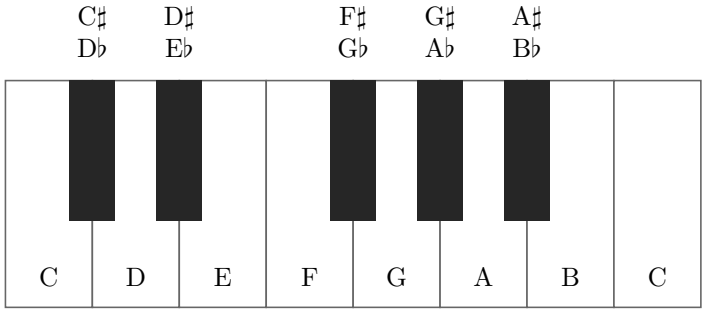


Figure 1.3: Each black key has two names: a sharp of the white key to its left, or a flat of the white key to its right.

A third sign, the **natural** (written ♮), cancels a sharp or flat. F♮ means plain F, the white key.

Try it at the keyboard

Put a finger on F. Move one key right: that is F♯. Now put a finger on G and move one key left: that is G♭. You are on the same black key both times.

1.8 A first look at intervals

The distance between two notes is called an **interval**. You measure it the way you measured steps: count the half steps from the lower note to the higher one. Put the lower note at zero, and count one for each key you move to.

For example, from C up to E: C♯ is 1, D is 2, D♯ is 3 and E is 4. So C to E is 4 half steps.

Some intervals have names, and each has its own sound. You can play the two notes one after the other, or together. Here are a few to listen to. You do not need to memorize them yet; Chapter 3 builds chords from the thirds.

Half steps	Name	Example	How it sounds together
1	half step	C to C♯	a rub, tense
2	whole step	C to D	a gentle rub
3	minor third	C to E♭	soft, a little sad
4	major third	C to E	bright, sweet
7	fifth	C to G	open, strong
12	octave	C to C	almost one note

Listen to each one, first apart and then together:

- Half step. The two notes rub against each other.
- Whole step.
- Minor third.
- Major third. The notes blend smoothly.

- Fifth. Even smoother and more open.
- Octave. The two notes nearly merge.

Intervals work the same everywhere on the keyboard. A major third is always 4 half steps, starting on any key. From D, count 4 half steps and you land on F♯. Hear C to E, then D to F♯: the same distance, the same sound, only higher.

For the mathematically curious

A note is a vibration. The A above middle C makes the air vibrate 440 times per second; we say its frequency is 440 hertz (Hz).

Going up an octave doubles the frequency: A5 is 880 Hz and A3 is 220 Hz. When one frequency is exactly twice the other, the waves line up every second vibration of the higher note, which is why octaves blend so well.

Other smooth intervals come from other simple ratios. A fifth is close to 3:2 (three vibrations of the top note for every two of the bottom), and a major third is close to 5:4. The half step, close to 16:15, has no simple pattern, and we hear it as a rub.

A modern piano splits each octave into 12 equal steps. Each half step multiplies the frequency by the same number r, and twelve of them make an octave, so $r^{12} = 2$ and $r = 2^{1/12} \approx 1.0595$. A fifth is then $2^{7/12} \approx 1.498$, very close to $3/2$. This system is called equal temperament.

Because the names repeat every 12 half steps, the twelve note names behave like the numbers on a clock. Put C at 12 o'clock, C♯ at 1, D at 2, and so on. Counting up 4 half steps from A (at 9 o'clock) lands on C♯ (1 o'clock), in the same way that 4 hours after 9 o'clock is 1 o'clock. Mathematicians call this arithmetic modulo 12.

What to remember

- A *note* is one musical sound; its *pitch* is how high or low it is. Left is lower, right is higher.
- The black keys come in groups of two and three. C is directly left of each group of two; F is directly left of each group of three.
- The white keys are named C, D, E, F, G, A, B, and then the letters repeat.
- An *octave* is 12 keys up, to the next note with the same name. It sounds like the same note, higher. Middle C is C4; the next C up is C5.
- A *half step* is one key to the very next key. A *whole step* is two half steps. Count every key, black and white. E–F and B–C are half steps.
- A *sharp* (♯) is one key higher; a *flat* (♭) is one key lower. Each black key has two names, such as C♯ and D♭.
- An *interval* is the distance between two notes, counted in half steps: a major third is 4, a fifth is 7, an octave is 12.

Exercises

Exercise 1.1. Name the white key that sits (a) between the two black keys of a group of two, (b) directly to the right of a group of three, (c) directly to the left of a group of three.

Answer

(a) D. The group of two sits around D: C is on its left, D in the middle, E on its right. (b) B. The group of three has F, G, A and B around it, and B is the white key on its right. (c) F, our second landmark.

Exercise 1.2. Start on B and go (a) one half step up, (b) one whole step up. Then start on F and do the same.

Answer

From B: (a) a half step up is C, a white key, because there is no black key between B and C. (b) A whole step up is two half steps, B to C to C♯, so you land on the black key C♯ (also called D♭). From F: (a) a half step up is the black key F♯. (b) A whole step up skips F♯ and lands on G.

Exercise 1.3. Give both names of (a) the black key between C and D, (b) the middle black key in a group of three. (c) Which white key is F♭?

Answer

(a) C♯ or D♭. (b) The middle black key of the three sits between G and A, so it is G♯ or A♭. (c) F♭ is one key to the left of F. There is no black key there, so it is the white key E.

Exercise 1.4. Count the half steps in each interval, going up: (a) C to G, (b) D to F♯, (c) A to C. Which ones are named in the table?

Answer

(a) C♯ 1, D 2, D♯ 3, E 4, F 5, F♯ 6, G 7: seven half steps, a fifth. (b) D♯ 1, E 2, F 3, F♯ 4: four half steps, a major third. (c) A♯ 1, B 2, C 3: three half steps, a minor third. All three are in the table.

Exercise 1.5. (a) What is the C one octave above middle C called? (b) What is the octave number of the B directly below middle C? (c) What is the A directly above middle C called?

Answer

(a) C5. (b) B3: octave numbers change at C, so the B directly below C4 still belongs to octave 3. (c) A4. It is the note that vibrates 440 times per second, used to tune orchestras.

Chapter 2

Scales and Keys

Most songs do not use all twelve notes of the octave equally. They pick seven of them and stay mostly with those. This chapter shows which seven, and why that choice makes one note feel like home.

You will build the major scale from one short pattern, start it on any key, and see what musicians mean by "the key of G". You will also meet the minor scale, a five-note scale you can improvise with today, and a map of how all the keys fit together.

2.1 What a scale is

A **scale** is a set of notes played in order, going up or down, like the rungs of a ladder. Different scales choose different notes from the twelve piano keys in an octave.

If you play every key from C up to the next C, black and white, you get the **chromatic scale**: twelve half steps in a row. All its steps are the same size, so no note stands out. It sounds like it could go on forever.

Now play only the white keys from C up to C. You may know it as do, re, mi, fa, sol, la, ti, do. This time the last C sounds like arriving home.

Try stopping one note early, on B. It feels unfinished, as if the music is waiting for C. The home note of a scale is called the **tonic**. Here the tonic is C, and this scale is called the C major scale.

2.2 The major scale pattern

Why does the C major scale have a clear home? Because its steps are not all the same size. Count the half steps between each pair of neighboring white keys from C to C, as in Chapter 1:

- C to D, D to E: whole steps.
- E to F: a half step (no black key between them).
- F to G, G to A, A to B: whole steps.

- B to C: a half step.

Write W for a whole step and H for a half step. The pattern of the major scale is:

W W H W W W H

Figure 2.1 shows it on the keyboard.

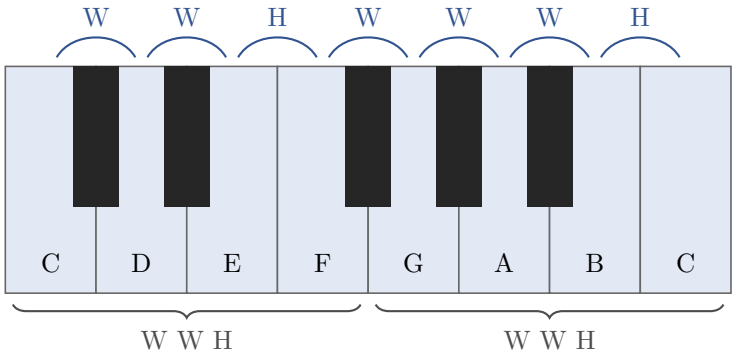


Figure 2.1: The C major scale uses only white keys. Its steps follow the pattern W W H W W W H. The two halves, C to F and G to C, have the same shape.

There is an easy way to remember the pattern. Split the eight notes into two halves of four notes: C–D–E–F and G–A–B–C. Each half has the same shape, W W H. A whole step, F to G, joins them.

So a major scale is: W W H, then one W to cross over, then W W H again. Hear the two halves.

This pattern is all you need. Start it on any key and you get a major scale with that key as its tonic. C major is the only one that uses all white keys, because its two half steps fall exactly on E–F and B–C.

2.3 Building major scales on other notes

G major

Start on G and follow the pattern.

- First half, W W H: G, then a whole step to A, a whole step to B, and a half step to C.
- Cross over with a whole step: C to D.
- Second half, W W H: D, a whole step to E, then a whole step from E. E to F is only a half step, so a whole step from E skips F and lands on the black key F♯. Then a half step from F♯ brings you home to G.

The G major scale is G, A, B, C, D, E, F♯ and G. It has one black key, F♯. Hear what happens if you play a plain F by mistake: the top no longer leads home so clearly.

F major

Start on F.

- First half, W W H: F, a whole step to G, a whole step to A, then a half step up from A. One key right of A is the black key we call B♭.
- Cross over with a whole step: B♭ to C.
- Second half, W W H: C, D, E and F, all white keys.

The F major scale is F, G, A, B♭, C, D, E and F. It has one black key, B♭. Figure 2.2 shows G major and F major side by side.

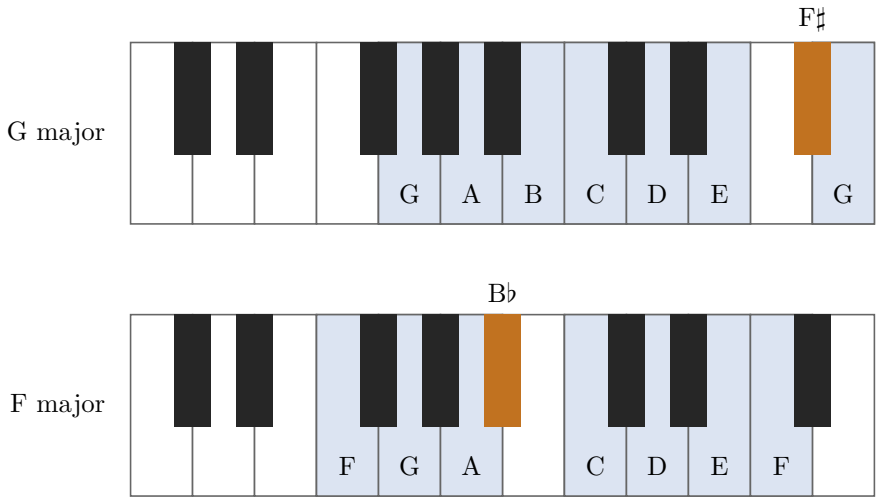


Figure 2.2: G major needs one black key, F♯. F major needs one black key, B♭. All their other notes are white keys.

D major

Start on D: whole step to E, whole step to F♯, half step to G. Cross over: G to A. Then whole step to B, whole step to C♯, half step to D.

The D major scale is D, E, F♯, G, A, B, C♯ and D. It has two black keys: F♯ and C♯.

Try it at the keyboard

Pick a starting key, say A, and build its major scale with the pattern W W H W W W H, counting every key. Check by ear: the last note should feel like home. (Answer: A, B, C♯, D, E, F♯, G♯, A.)

2.4 One letter for each note

Why do we call the black key in F major B♭ and not A♯? They are the same key. The answer is a simple naming rule.

A major scale uses each of the seven letters exactly once, in order. F major uses F, G, A, B, C, D, E: every letter, none repeated. If we called the fourth note A♯, we would have two kinds of A and no B at all.

The same rule explains F♯ in G major. The scale needs a note between E and G, so it must be some kind of F. That is F♯, not G♭.

So to name the notes of a scale, write the seven letters in order from the tonic first. Then add a sharp or flat to any letter that needs one to fit the pattern.

2.5 What "the key of G" means

The word "key" has two meanings in music. One is a piano key, the thing you press. The other is a musical **key**, which describes a whole piece of music.

A song in the key of G is built mostly from the notes of the G major scale, and G is its home note. Its melody and chords use G, A, B, C, D, E and F♯. It tends to end on G, because G sounds like home.

So when a musician says "this song is in G", you know two things at once: which seven notes to expect, and which note is home. In the key of G you play F♯ instead of F, almost always.

Each key has its own number of sharps or flats. The key of C has none, G has one sharp (F♯), D has two sharps (F♯ and C♯), and F has one flat (B♭). Sheet music shows these at the start of every line; Chapter 6 shows how this looks on paper.

2.6 Playing the C major scale

You have five fingers, but a scale has eight notes. Pianists solve this by tucking the thumb under the hand partway up the scale, so the hand moves along smoothly.

Right hand

Put your right thumb on middle C.

- Going up: play C, D, E with fingers 1–2–3. While finger 3 holds E, pass your thumb under your hand to play F. Then play F, G, A, B, C with fingers 1–2–3–4–5.
- Going down: play C, B, A, G, F with fingers 5–4–3–2–1. Then cross finger 3 over your thumb to play E, and finish E, D, C with 3–2–1.

So the fingers going up are 1–2–3–1–2–3–4–5, and going down they are the same in reverse. Hear it slowly.

Left hand

Put your left little finger on the C below middle C, which is C3.

- Going up: play C, D, E, F, G with fingers 5–4–3–2–1. Then cross finger 3 over your thumb to play A, and finish A, B, C with 3–2–1.
- Going down: play C, B, A with fingers 1–2–3. Pass your thumb under to play G, then play G, F, E, D, C with 1–2–3–4–5.

Hear the left hand going up and down.

Try it at the keyboard

Play the right hand alone, very slowly, until the thumb passes under without a bump. Keep your wrist level and let the thumb slide under a relaxed, rounded hand. Then try the left hand. Speed does not matter yet; evenness does.

2.7 The natural minor scale

The major scale sounds bright. The **minor scale** has a darker, more serious sound. You do not need a new pattern to find it: every major scale hides a minor scale inside it.

Play the white keys again, but start and end on A instead of C: A, B, C, D, E, F, G and A. These are the same seven notes as C major, but now A is home. This is the **A natural minor scale** ("natural" here means the plain, basic form; it has nothing to do with the natural sign).

A minor is called the **relative minor** of C major, because the two scales share all their notes. To find the relative minor of any major key, start on the sixth note of its major scale. Or count down 3 half steps from the tonic: from C, down to B, B♭ and A.

Hear C major and then A minor. Same keys, different home, different mood.

Why it sounds darker

Your ear measures every note from the tonic. In C major, the third note, E, is 4 half steps above C: a major third. In A minor, the third note, C, is only 3 half steps above A: a minor third. That smaller distance gives minor its darker color.

Hear C major and C minor, both with C as home. The C minor scale is C, D, E♭, F, G, A♭, B♭ and C. Its notes are the same as E♭ major, because C is 3 half steps below E♭.

Two more minor scales

The relative minor of G major is E minor: E, F♯, G, A, B, C, D and E. The relative minor of F major is D minor: D, E, F, G, A, B♭, C and D.

Composers often raise the seventh note of a minor scale by a half step, so it leans more strongly into the tonic. In A minor that turns G into G♯. The result is called the **harmonic minor scale**.

2.8 The pentatonic scale

A **pentatonic scale** has only five notes. Take C major and leave out F and B, the two notes that make half steps. Five notes remain: C, D, E, G and A. This is the C major pentatonic scale.

With no half steps, none of its notes make the tense rub of a half step. Almost any order of these five notes sounds like a pleasant tune. That is why pentatonic scales appear in folk songs all over the world.

The five black keys form a pentatonic scale too: G♭, A♭, B♭, D♭ and E♭ (Figure 2.3). Hear them. Started on G♭, they make the G♭ major pentatonic scale.

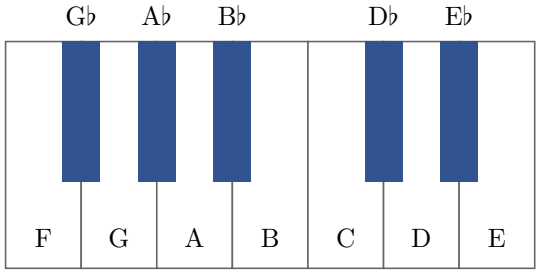


Figure 2.3: The five black keys make a pentatonic scale: G♭, A♭, B♭, D♭ and E♭.

Try it at the keyboard

Improvise on the black keys. With your right hand, play short, slow phrases using only black keys, in any order, with small pauses between them. With your left hand, hold a low G♭ now and then. End a phrase on G♭ and it will sound finished. Here is one idea to start from.

2.9 The circle of fifths

You have seen that C major has no sharps, G major has one, and D major has two. There is a pattern here. G is a fifth above C (7 half steps), and D is a fifth above G.

Each time you move up a fifth, the new major scale keeps six notes of the old one and changes only one: one note goes up a half step and becomes a sharp. From C to G, F becomes F♯. From G to D, C becomes C♯ and F♯ stays.

Keep going up by fifths: C, G, D, A, E, B, F♯. Each step adds one more sharp. After twelve steps you are back at C, having visited all twelve keys (past F♯, musicians switch to flat names). Laid out in a circle, as in Figure 2.4, this is the **circle of fifths**.

Going the other way, counterclockwise, each step moves down a fifth and adds one flat: C, F, B♭, E♭, A♭, D♭, G♭. At the bottom of the circle the two directions meet: F♯ major and G♭ major use exactly the same piano keys, named differently.

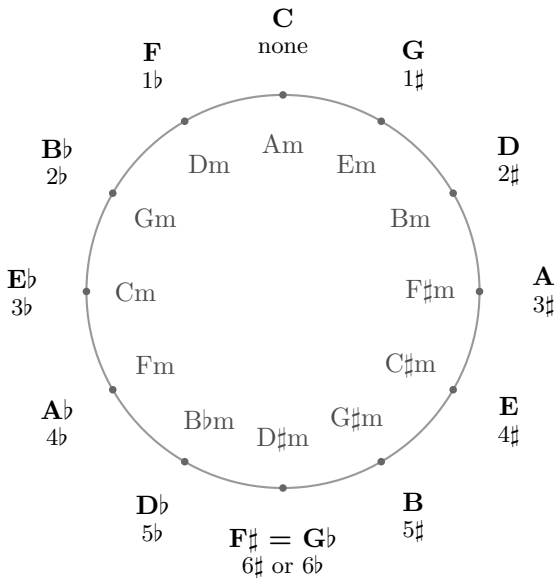


Figure 2.4: The circle of fifths. Going clockwise, each key is a fifth higher and has one more sharp (or one fewer flat). Inside each major key is its relative minor.

A map of neighbors

Think of the circle as a map of a town where neighbors are friendly. Two keys next to each other share six of their seven notes. That is why music often moves to a neighboring key: it sounds like a short trip, not a jump.

Keys far apart on the circle, such as C and F♯, share very few notes. Moving between them sounds much more surprising.

The relative minors sit inside the circle, next to their major partners. A minor is inside C, E minor inside G, and so on. A relative minor has the same sharps or flats as its major partner.

Which sharps, which flats

Sharps always appear in the same order: F♯, C♯, G♯, D♯, A♯, E♯, B♯. A key with two sharps has the first two, F♯ and C♯. A key with three sharps has F♯, C♯ and G♯, and so on.

Flats appear in the reverse order: B♭, E♭, A♭, D♭, G♭, C♭, F♭. A key with three flats has B♭, E♭ and A♭.

Two handy facts help you check. In a sharp key, the newest sharp is the seventh note of the scale, a half step below the tonic: in D major, C♯ is a half step below D. In a flat key, the newest flat is the fourth note: in F major, B♭ is the fourth note.

For the mathematically curious

Why does stepping by fifths visit all twelve keys before coming back? Number the keys 0 to 11 around the clock (C = 0, C♯ = 1, and so on up to B = 11). A fifth adds 7, and we wrap around after 12, as on a clock.

Starting at 0, the steps give 0, 7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, and then 0 again: all twelve numbers. This happens because 7 and 12 have no common factor except 1; in symbols, $\gcd(7, 12) = 1$.
Compare stepping by 3 (a minor third): 0, 3, 6, 9, 0. It returns after only four steps, because 3 divides 12. Stepping by 4 (a major third) visits only three keys. Only steps of 1, 5, 7 and 11 half steps visit every key, and 5 and 7 give the two directions around the circle of fifths.

What to remember

- A *scale* is a set of notes in order. Its home note is the *tonic*.
- The *major scale* follows the pattern W W H W W W H (W = whole step, H = half step): two matching halves, W W H, joined by a W.
- A major scale uses each letter once. That rule decides whether a black key is called a sharp or a flat.
- A piece in a musical *key*, such as the key of G, uses the notes of that major scale and treats its tonic as home.
- C major scale fingering, right hand going up: 1–2–3, thumb under, 1–2–3–4–5.
- The *relative minor* starts on the sixth note of a major scale (3 half steps below the tonic) and uses the same notes. Minor sounds darker.
- The *pentatonic scale* has five notes and no half steps. The black keys form one.
- On the *circle of fifths*, each step clockwise adds a sharp and each step counterclockwise adds a flat. Neighbors share six of seven notes.

Exercises

Exercise 2.1. Build the D major scale with the pattern W W H W W W H. Which notes are black keys?

Answer

Start on D. Whole step to E, whole step to F♯, half step to G. Whole step to A. Whole step to B, whole step to C♯, half step to D. The scale is D, E, F♯, G, A, B, C♯, D, with two black keys: F♯ and C♯.

Exercise 2.2. Build the B♭ major scale, using each letter once. How many flats does it have?

Answer

Start on B♭. Whole step to C (skipping B), whole step to D, half step to E♭. Whole step to F. Whole step to G, whole step to A, half step to B♭. The scale is B♭, C, D, E♭, F, G, A, B♭. It has two flats, B♭ and E♭. We call the fourth note E♭ and not D♯ because the scale already has a D and needs an E.

Exercise 2.3. Find the relative minor of (a) G major, (b) F major, (c) D major. Write out the notes of (c).

Answer

Count down 3 half steps from each tonic, or take the sixth note of the major scale. (a) E minor. (b) D minor. (c) From D, down 3 half steps: C♯, C, B. So B minor. It uses the notes of D major starting on B: B, C♯, D, E, F♯, G, A, B.

Exercise 2.4. Use the order of sharps and flats. (a) A major has three sharps; which ones? (b) E♭ major has three flats; which ones?

Answer

(a) The first three sharps: F♯, C♯ and G♯. Check: G♯ is a half step below A, the tonic. (b) The first three flats: B♭, E♭ and A♭. Check: A♭ is the fourth note of E♭ major (E♭, F, G, A♭).

Exercise 2.5. Which single note is different between (a) C major and G major, (b) C major and F major? Where are these keys on the circle?

Answer

(a) F in C major becomes F♯ in G major. (b) B in C major becomes B♭ in F major. G and F are the two neighbors of C on the circle, one on each side, which is why each differs from C by one note.

Part II

Chords Under Your Hands

Chapter 3

Chords: Notes Played Together

So far you have played notes one after another. A tune you can hum works that way: one note, then the next. A line of notes heard one after another is called a **melody**.

Now press three keys at the same moment. The notes blend into one fuller sound. Three or more notes played together make a **chord**. The sound of chords supporting a melody is called **harmony**.

Think of a singer in a room. The melody is the voice. The harmony is the lighting: the voice stays the same, but the mood of the room changes. Hear a short melody alone, and then the same melody with chords under it.

In this chapter you will build chords with one hand. You will hear why some chords sound bright and others sound sad. And you will meet the small family of chords that most songs use.

3.1 Your first chord: C major

Put your right thumb on middle C (C4, from Chapter 1). This is where the chord starts, and it gives the chord its name. The note a chord is built on is called its **root**.

Now count up 4 keys from C. Count every key, black or white, and do not count the key you start on. You pass C♯, D and D♯, and land on E. Counting keys like this is counting half steps, as in Chapter 1.

From E, count up 3 more keys: F, F♯, G. You land on G. Play C, E and G together. This is the **C major chord**. Its chord symbol is the letter C on its own.

Use fingers 1, 3 and 5: thumb, middle finger and little finger. Fingers 2 and 4 float above the keys they skip. Hear the three notes one at a time, then together.

A chord of three notes built this way is called a **triad** ("tri" means three). Each of its notes has a name:

- the root is the bottom note, C, which names the chord.
- the **third** is the middle note, E. It is three letters up from the root: C, D, E. (C up to E is the major third from Chapter 1.)

- the **fifth** is the top note, G. It is five letters up: C, D, E, F, G. (C up to G is the fifth from Chapter 1.)

Common trap

When you count 4 keys, count the black keys too. If you count only white keys, you land on G instead of E. A good check: the major chord on C uses only white keys, and your hand looks like "play one, skip one, play one, skip one, play one".

3.2 Major and minor: one key makes the difference

Keep your thumb on C and your little finger on G. Now move only your middle finger one key to the left, from E down to the black key E♭. Play C, E♭ and G together.

This is the **C minor chord**. Its symbol is Cm; the small m means minor. Play C major and then C minor, and listen. Most people hear major as bright and settled, and minor as darker or sadder.

The root did not change, and neither did the top note. Only the middle note moved, by one key. That gives two short recipes, which work from any root:

- Major: from the root, count up 4 keys, then 3 more. We will write this as "4 then 3".
- Minor: from the root, count up 3 keys, then 4 more: "3 then 4".

In both, the top note is 7 keys above the root. Figure 3.1 shows the two side by side.

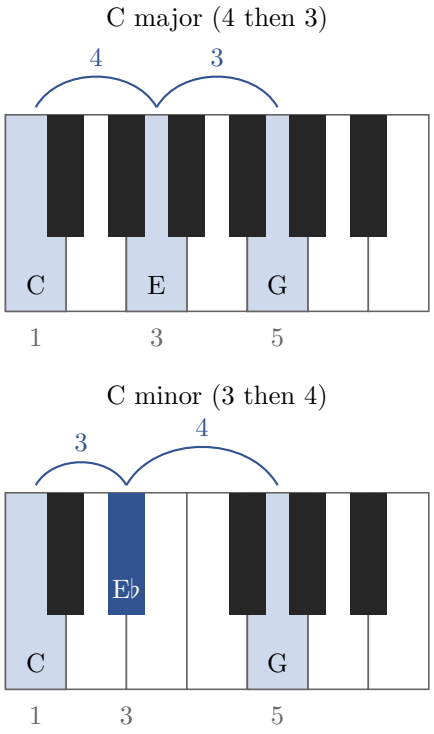


Figure 3.1: C major and C minor share C and G. Only the middle note moves, from E down to E♭. The small numbers under the keys are the fingers to use.

Try the recipe on D. Count 4 keys up from D: D♯, E, F, F♯. Then 3 more: G, G♯, A. So D major is D, F♯ and A. Move the middle finger down one key to F, and you have D minor, Dm: D, F and A.

Now A. A major is A, C♯ and E, and A minor, Am, is A, C and E.

Notice that a triad always skips a letter between its notes: D, (E), F♯, (G), A. That is why we call the middle note of D major F♯ and not G♭, even though it is the same key. As with scales in Chapter 2, the letter name follows the pattern.

Try it at the keyboard

Build G major and G minor, then E major and E minor, by counting. Play each pair and listen for the change of mood. You should find G, B, D and G, B♭, D, then E, G♯, B and E, G, B.

3.3 A shortcut on the white keys

Counting works from any key. On the white keys there is a quicker way to find a chord: play a white key, skip a white key, play, skip, play. From C that gives C, E and G.

Try that shape starting on each white key in turn. Some of the chords sound bright and some sound dark:

- starting on C, F or G gives a major chord: C (C, E, G), F (F, A, C) and G (G, B, D);
- starting on D, E or A gives a minor chord: Dm (D, F, A), Em (E, G, B) and Am (A, C, E);
- starting on B gives B, D and F, which is neither. Count it: 3 keys, then 3 more. We meet this chord in the next section.

Why does one hand shape give different chords? Because the white keys are not evenly spaced. There is no black key between E and F, or between B and C. So the same "skip a white key" gap is sometimes 4 keys and sometimes 3. When in doubt, count.

3.4 Two more sounds: sus and diminished

Play C major, then lift your middle finger off E and put finger 4 on F. You now hold C, F and G. This is **Csus4** (said "C sus four"). "Sus" is short for suspended: the middle note is hung up one key too high, so the chord is neither major nor minor.

A sus chord sounds open and unfinished. It wants to fall back to the major chord. Hear the pull from Csus4 to C. You can also use finger 2 on D instead: C, D and G is **Csus2**.

The chord you found on B, with B, D and F, is a **diminished** chord. Its recipe is 3 then 3. The top note is only 6 keys above the root instead of 7, and the chord sounds tense. It is written Bdim, or B° with a small circle. Hear it lean into C major.

Here are all the recipes so far, with C as the root. You do not need to memorize the table; it is here to look back at.

Chord	Symbol	Count up	Notes from C
Major	C	4 then 3	C, E, G
Minor	Cm	3 then 4	C, E♭, G
Diminished	Cdim	3 then 3	C, E♭, G♭
Sus4	Csus4	5 then 2	C, F, G
Sus2	Csus2	2 then 5	C, D, G

3.5 Three hand shapes for most major chords

Counting is reliable, but pianists do not count while they play. They remember how each chord looks and feels under the hand. Nine of the twelve major chords fall into three shapes, shown in Figure 3.2:

- White, white, white: C, F and G major.
- White, black, white: D, E and A major. The middle finger sits up on a black key.
- Black, white, black: D♭, E♭ and A♭ major. Thumb and little finger sit on black keys, with the middle finger down on a white key. The notes are D♭, F, A♭; E♭, G, B♭; and A♭, C, E♭.

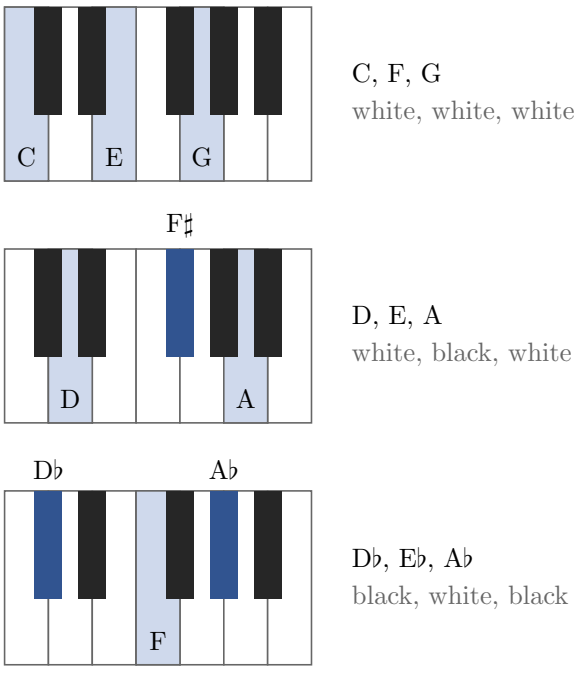


Figure 3.2: Three shapes cover nine major chords. Each keyboard shows one example: C major, D major and D♭ major.

That leaves three major chords, and each is easy to remember on its own:

- G♭ major (also called F♯ major) is all black keys: G♭, B♭, D♭.
- B♭ major is black, white, white: B♭, D, F.
- B major is white, black, black: B, D♯, F♯.

Intuition

Minor chords swap the first two shapes. Lower the middle note of C, F or G major and you get white, black, white: Cm is C, E♭, G. Lower the middle note of D, E or A major and you get three white keys: Dm, Em and Am. So for these six roots, one look tells you the shape of both the major and the minor chord.

3.6 The chords that belong to a key

Recall from Chapter 2 that a song "in the key of C" uses mostly the notes of the C major scale, the white keys, and feels at home on C. Which chords fit with such a song? The ones built only from those same notes.

To find them, put fingers 1, 3 and 5 on C, E and G. Keep the "play, skip, play, skip, play" shape and move your whole hand up one white key at a time. You get seven chords, one on each note of the scale. These are the chords of the key of C.

Musicians number these chords by the scale note they start on, using Roman numerals: I is 1, II is 2, III is 3, IV is 4, V is 5, VI is 6 and VII is 7. A capital numeral means a major chord and a small one means a minor chord. The little circle on vii° means diminished.

Numeral	Built on	Chord	Notes	Sound
I	1st note	C	C, E, G	major
ii	2nd note	Dm	D, F, A	minor
iii	3rd note	Em	E, G, B	minor
IV	4th note	F	F, A, C	major
V	5th note	G	G, B, D	major
vi	6th note	Am	A, C, E	minor
vii°	7th note	Bdim	B, D, F	diminished

Why use numbers instead of names? Every major scale has the same pattern of whole and half steps (Chapter 2). So every major key gives the same pattern of chords: major on 1, 4 and 5, minor on 2, 3 and 6, and diminished on 7.

For example, the key of G uses the G major scale: G, A, B, C, D, E, F♯. Its chords are G, Am, Bm, C, D, Em and F♯dim. The numbers stay the same: in G, the V chord is D. A song described as "I, IV, V" can be played in any key once you know that key's chords.

Home, going away, and coming home

The chords of a key have different jobs, a bit like places in a story:

- I is home. It sounds settled. Most songs end on it.
- IV and ii step away from home. They give a feeling of setting out.
- V is tension. It pulls strongly back toward I.

Hear the pull of V. In C, play C, F, G and stop: the music feels unfinished. Now let it come home to C.

Part of the pull comes from one note. The G chord contains B, which is only one key below C. Your ear expects that B to rise to C, and when it does, you feel arrival. The vi chord, Am, is home's quieter cousin: it shares two notes, C and E, with the C chord.

3.7 Famous chord progressions

A **chord progression** is a chain of chords that a song repeats, usually with a steady number of counts on each chord. A few progressions turn up in thousands of songs. Here are four, in the key of C. Each button plays the chords once, two counts each, and then lands on the first chord again.

- I–V–vi–IV: C, G, Am, F. One of the most common progressions in pop music.
- vi–IV–I–V: Am, F, C, G. The same four chords, starting on the minor one, so it feels more wistful.
- I–vi–IV–V: C, Am, F, G. The sound of many 1950s songs.
- ii–V–I: Dm, G, C. A short trip home, used everywhere in jazz.

To move a progression to another key, keep the numerals and read the chords from the new key. In the key of G, I is G, ii is Am, IV is C, V is D and vi is Em:

Numerals	In C	In G
I–V–vi–IV	C, G, Am, F	G, D, Em, C
vi–IV–I–V	Am, F, C, G	Em, C, G, D
I–vi–IV–V	C, Am, F, G	G, Em, C, D
ii–V–I	Dm, G, C	Am, D, G

Hear I–V–vi–IV in G and ii–V–I in G.

Try it at the keyboard

Play I–V–vi–IV in C with your right hand: C, G, Am, F. Count 1, 2, 3, 4 slowly and steadily, and change chord each time you get back to 1. Your hand will jump around for now. In Chapter 4 you will learn to make these changes with almost no movement.

3.8 Seventh chords: adding a fourth note

Keep going with the "play, skip, play" pattern one more step. From C you get C, E, G and then B. A four-note chord like this is a **seventh chord**, because its top note is seven letters above the root: C, D, E, F, G, A, B.

There are three seventh chords to know first. Their symbols tell you which one you have:

- Cmaj7 ("C major seven") is C, E, G, B. It sounds soft and dreamy.

- Dm7 ("D minor seven") is D, F, A, C. It sounds mellow.
- G7 ("G seven") is G, B, D, F. It has a strong pull toward C. Inside it, B wants to rise one key to C, and F wants to fall one key to E.

Common trap

C7 and Cmaj7 are different chords. C7 is C, E, G, B♭: its top note is the black key B♭. Cmaj7 is C, E, G, B. A plain 7 always means the lower top note, which sounds bluesy and restless; maj7 means the higher one. Compare them.

Four notes in one hand can be a stretch. Here is a trick: play the root low with your left hand, and the other three notes with your right hand. Those top three notes always form a triad you already know:

- Cmaj7 is C under E, G, B. The right hand plays Em.
- Dm7 is D under F, A, C. The right hand plays F major.
- G7 is G under B, D, F. The right hand plays Bdim.
- Am7 (A, C, E, G) is A under C, E, G. The right hand plays C major.

Work it yourself

Play ii–V–I in C with seventh chords: Dm7, G7, Cmaj7. Count 4 slow counts on each chord. Left hand: D3, then G2, then C3, one finger at a time. Right hand: F, A, C, then F, G, B, then E, G, B, all a little above middle C. (In G7 the right hand leaves out D; you will not miss it.) Your right hand barely moves.

For the mathematically curious

Place the twelve notes around a clock face, C at 12 o'clock, one note per hour: C♯ at 1, D at 2, and so on. Join the three notes of a major chord and you get a triangle whose sides span 4, 3 and 5 hours. Every major chord is the same triangle, turned to a new position. That is why one counting recipe works from every root.
A minor chord is the same triangle flipped over: 3, 4 and 5 hours. A diminished chord is 3, 3 and 6. The chord C, E, G♯ (4, 4 and 4) makes a perfect equilateral triangle; it is called augmented.

What to remember

- A *melody* is notes one after another; a *chord* is notes played together; *harmony* is the sound of chords under a melody.
- A *triad* has three notes: the *root* (which names it), the *third* and the *fifth*. Play it with fingers 1, 3 and 5.
- Major: count up 4 keys, then 3 more. Minor: 3, then 4. Only the middle note differs.
- Major sounds bright; minor sounds darker. Sus chords sound open and unfinished; diminished chords sound tense.
- The chords of a major key, numbered by Roman numerals, are always I, ii, iii, IV, V, vi,

vii°. In C: C, Dm, Em, F, G, Am, Bdim.

- I is home, IV steps away, and V pulls back home. A *chord progression* is a chain of chords a song repeats, such as I–V–vi–IV.
- A *seventh chord* adds a fourth note. G7 is not Gmaj7. Play the root in the left hand and a familiar triad in the right.

Exercises

Exercise 3.1. Build E major and E♭ major by counting 4 then 3. Which of the three shapes does each belong to? Then lower the middle note of each to get Em and E♭m.

Answer

From E, count 4 keys: F, F♯, G, G♯. Then 3 more: A, A♯, B. So E major is E, G♯, B. That is white, black, white.
From E♭, count 4 keys: E, F, G♭, G. Then 3 more: A♭, A, B♭. So E♭ major is E♭, G, B♭. That is black, white, black.
Lowering the middle notes gives Em: E, G, B, all white keys. And E♭m: E♭, G♭, B♭, all black keys. We call the middle note G♭ rather than F♯ so that the letters skip one each time: E, G, B.

Exercise 3.2. Spell Dsus4 (5 then 2 from D) and Dsus2 (2 then 5 from D). Compare them with D major.

Answer

Dsus4: from D, count 5 keys to G, then 2 more to A. So Dsus4 is D, G, A. Dsus2: count 2 keys to E, then 5 more to A. So Dsus2 is D, E, A.
D major is D, F♯, A. All three share D and A. Only the middle note changes: G for sus4, F♯ for major, E for sus2. Hear Dsus4 settle into D major.

Exercise 3.3. The F major scale is F, G, A, B♭, C, D, E. List the seven chords of the key of F, with their Roman numerals. Then play I–V–vi–IV in F.

Answer

Use the "play, skip, play, skip, play" shape on the notes of the F major scale. You get: I is F (F, A, C); ii is Gm (G, B♭, D); iii is Am (A, C, E); IV is B♭ (B♭, D, F); V is C (C, E, G); vi is Dm (D, F, A); vii° is Edim (E, G, B♭).
So I–V–vi–IV in F is F, C, Dm, B♭.

Exercise 3.4. A song in the key of G uses only three chords: G, C and D. What are their Roman numerals? Which three chords would the same song use in the key of D? (The D major scale is D, E, F♯, G, A, B, C♯.)

Answer

In G, the chord on the 1st note is G, the 4th note is C, and the 5th note is D. So the song uses I, IV and V. In the key of D, the 1st, 4th and 5th notes are D, G and A, so the song uses D, G and A major.

Exercise 3.5. Em7 is E, G, B, D. Which triad does the right hand play over a left-hand E? And D7 is D, F♯, A, C: which two of its notes move by one key to reach the G chord (G, B, D)?

Answer

Take away the root E and you are left with G, B, D: a G major triad. So Em7 is E in the left hand under G major in the right hand.
In D7, F♯ rises one key to G, and C falls one key to B. That is the same pull you met in G7 going to C, moved to a new key.

Chapter 4

Inversions and Smooth Chord Changes

Play C, F, G and back to C with the chord shapes from Chapter 3. Your whole hand jumps up to F, then a little higher to G, then all the way back down. Each jump takes time, and the music stumbles while your hand is in the air.

An experienced pianist plays the same three chords with a hand that hardly moves. Listen to both: the chords are the same, but the second version flows. The trick is to rearrange the notes of each chord. This chapter shows you how, and then adds the left hand so that your playing sounds full.

4.1 Same notes, different bottom

C major is the three notes C, E and G. It does not matter which of them is at the bottom: any arrangement of C, E and G is still a C major chord. A chord played with a note other than the root at the bottom is called an **inversion**.

Start with C4, E4, G4. This is **root position**: the root is at the bottom. Now take the bottom note, C4, and move it up to the next C, C5. You get E4, G4, C5: the **first inversion**. Do it again: move E4 up to E5, and you get G4, C5, E5, the **second inversion**. Hear all three.

All three sound like C major. The top note changes, so each has a slightly different color, but none of them sounds like a new chord. Figure 4.1 shows the three positions and the fingers to use.

- Root position, C4–E4–G4: fingers 1–3–5.
- First inversion, E4–G4–C5: fingers 1–2–5. The index finger takes G, which leaves room for the bigger stretch from G up to C.
- Second inversion, G4–C5–E5: fingers 1–3–5. Here the big stretch is at the bottom, between thumb and middle finger.

Try it at the keyboard

Play the three positions of C major one after another, going up: C4–E4–G4, then E4–G4–C5, then G4–C5–E5, then C5–E5–G5. Then come back down. Say the name of each position as you play it.

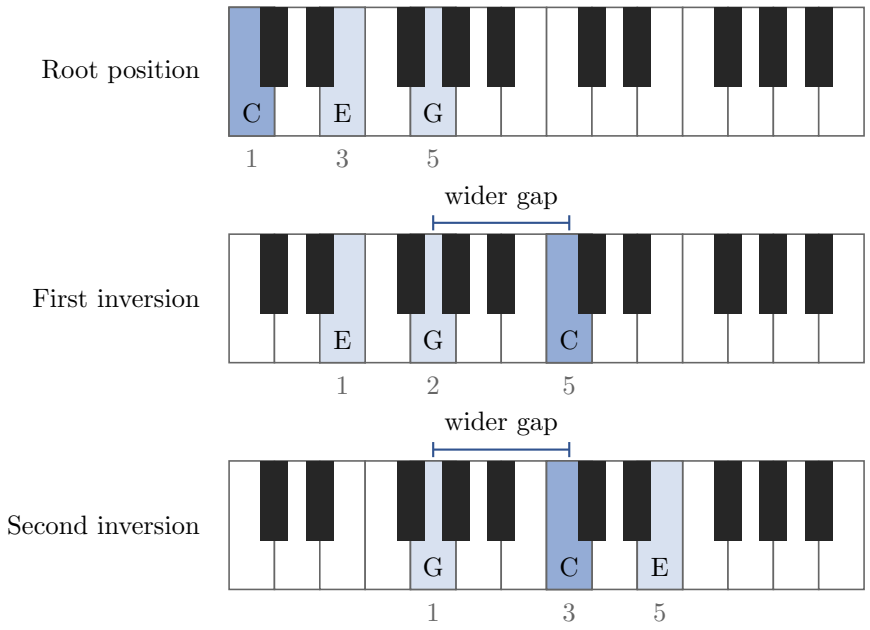


Figure 4.1: The three positions of C major. The root, C, is the darker key; in both inversions it sits directly above the wider gap. Small numbers are the right-hand fingers.

4.2 Finding the root of any chord

When a chord is inverted, its root is no longer at the bottom. So how do you name a chord you find under your fingers? Look at the two gaps between its three notes.

In root position both gaps are small: 3 or 4 keys each, and each skips one letter (C, E, G skips D and F). In an inversion, one gap is wider: 5 keys, skipping two letters. In E–G–C, the gap from G up to C skips A and B. Here is the rule:

> The note directly above the wider gap is the root.

Try it on A–D–F♯. From A up to D is 5 keys: the wider gap. From D up to F♯ is 4 keys. The note above the wider gap is D, so this is D major, in second inversion.

Try E–G–C. The wider gap is G up to C, so the root is C. The bottom note, E, is the third of C major, so this is first inversion; with the fifth at the bottom it would be second inversion. If neither gap is wider than 4 keys, the chord is in root position and its bottom note is the root.

4.3 Smooth chord changes: keep the shared note still

Two chords often share a note. C major (C, E, G) and F major (F, A, C) share C. C major and G major (G, B, D) share G. Pianists use this: they keep the shared note under the same finger, and move the other fingers to the nearest notes of the new chord.

Intuition

Think of the three notes of a chord as three singers in a choir. When the chord changes, no singer wants to leap. Each moves to the nearest note of the new chord, and anyone who already has a right note stays put.

Here is C to F, starting from C4–E4–G4 with fingers 1–3–5:

- the shared note is C, so your thumb stays on C4;
- E4 moves up one key to F4, and G4 moves up two keys to A4.

You now hold C4–F4–A4, which is F major in second inversion. Your hand did not travel at all.

Now C to G, again from C4–E4–G4:

- the shared note is G, so your little finger stays on G4;
- E4 moves down two keys to D4, and C4 moves down one key to B3.

You now hold B3–D4–G4, which is G major in first inversion, played with fingers 1–2–5.

Put it together: C, F, G, C. Compare the two versions from the start of the chapter. With root-position chords every finger moves at every change: added together, your fingers travel 42 keys. With the smooth version they travel 12. Figure 4.2 draws the two versions.

The same idea works for any progression. Here is I–V–vi–IV in C (C, G, Am, F) from Chapter 3, played smoothly:

- C: C4–E4–G4, fingers 1–3–5.
- G: B3–D4–G4, fingers 1–2–5. G stays.
- Am: C4–E4–A4, fingers 1–2–5. Every finger moves up a little.
- F: C4–F4–A4, fingers 1–3–5. C and A stay; only E moves up one key to F.

Hear it, ending back on C. Compare it with the jumpy version.

Common trap

Do not try to work out the inversion names while you play. Use the shared note and the shortest move instead. The name (first or second inversion) is useful for talking about a chord, not for finding it.

4.4 Two hands: a bass note below, a chord above

Your right hand can now change chords smoothly near middle C. But in C4–F4–A4 the lowest note is C, not F. How does a listener hear an F chord? The left hand tells them. It plays the root of the chord, low down. The lowest notes of a piece are called the **bass**.

The usual layout, shown in Figure 4.3:

- The left hand plays the root of each chord, low, between C2 and C3. It can play the root alone,

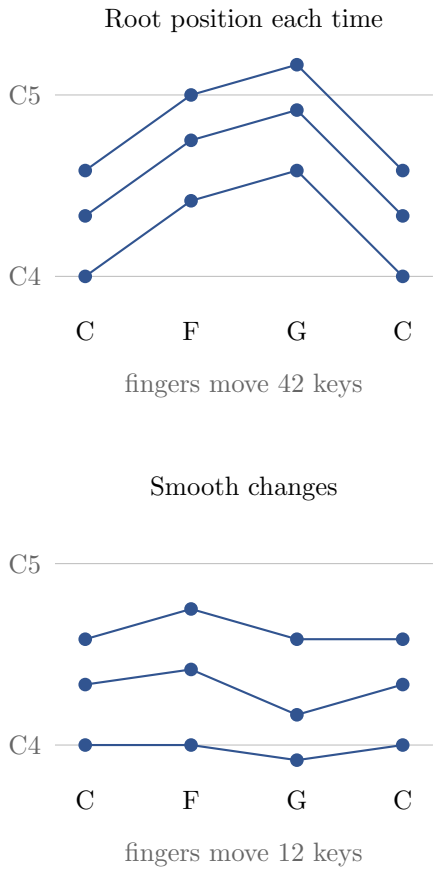


Figure 4.2: Each line follows one finger through C, F, G, C; higher means a higher key. At the top all three fingers jump; at the bottom they barely move.

the root with the note 7 keys above it (C2 and G2, called a fifth), or the root with the next root up (C2 and C3, called an octave).

- The right hand plays the chord near middle C, using smooth inversions.

Here is I–V–vi–IV with the left hand on C3, G2, A2 and F2 under the smooth right-hand chords.

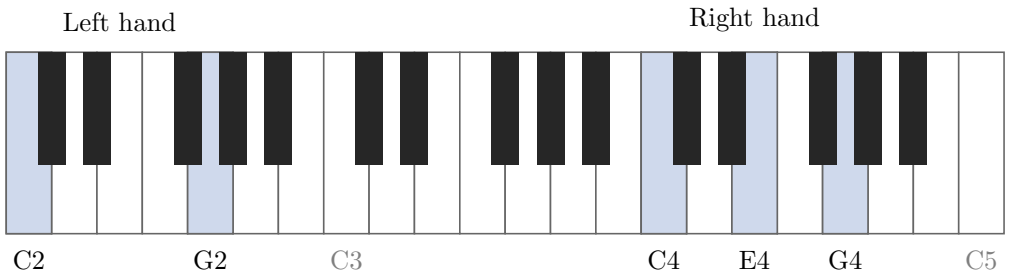


Figure 4.3: The left hand plays the root low (here C2 with G2). The right hand plays the chord near middle C.

Common trap

Do not play a full three-note chord low down in the left hand. Low notes close together sound thick and muddy. Compare a low C2–E2–G2 with a low C2–G2 under the same right hand. Below about C3, keep the left hand to single notes, fifths and octaves.

4.5 Three accompaniment patterns

Playing chords to support a melody or a singer is called **accompaniment**. Here are three patterns. Each one uses the smooth I–V–vi–IV from above, with 4 counts on each chord. Count 1, 2, 3, 4 steadily and change chord each time you return to 1. Rhythm gets a full chapter later (Chapter 5); for now, steady counting is all you need.

Block chords

Both hands play together on every count: left-hand root and right-hand chord, four times per chord. It is the plainest pattern, and the best one to start with.

Broken chords

The right hand plays the chord one note at a time instead of all at once. On count 1, the left hand plays the root together with the right hand's bottom note, and holds it down for all 4 counts. Then the right hand goes on: middle note on 2, top note on 3, middle note on 4. It sounds like a gentle rolling wave.

The pop bounce

The hands take turns. On 1, the left hand plays the low root and the right hand plays the chord, together. On 2, the right hand plays the chord alone. On 3, the left hand alone plays the note 7 keys above the root (with the thumb). On 4, the right hand plays the chord again.

Try it at the keyboard

Pick one pattern and play it on a single chord, C, until it feels easy. Then add the change to G, then Am, then F. Go slowly enough that you never have to stop. A steady slow pattern sounds much better than a fast one with gaps.

4.6 Slash chords and stepwise bass lines

Song sheets sometimes show chords like C/E or G/B. The slash does not mean a fraction. It means: play the chord on the left, with the note on the right in the bass. So C/E means a C major chord with E as the lowest note. The right hand plays C major and the left hand plays E.

Why would a song ask for this? To make the bass line move smoothly. Here is a famous kind of opening: C, G/B, Am, Am/G, F. Look at the left-hand notes: C3, B2, A2, G2, F2. The bass walks down the white keys one at a time, while the right hand plays smooth chords on top. A bass that moves to a neighboring note each time, like this, is called a **stepwise bass line**; musicians also say the bass "walks down".

Work it yourself

Put your left thumb on C3. Fingers 2, 3, 4 and 5 then sit on B2, A2, G2 and F2. Put your right hand on C4–E4–G4. Play four slow counts on each chord:

- C: left C3, right C4–E4–G4;
- G/B: left B2, right B3–D4–G4;
- Am: left A2, right C4–E4–A4;
- Am/G: left G2, right C4–E4–A4 (the right hand does not move);
- F: left F2, right C4–F4–A4.

When it feels comfortable, try it with your eyes closed, and let the shared notes guide your hand.

What to remember

- An *inversion* is a chord with a note other than the root at the bottom. It still has the same name.
- C major has three positions: root position C–E–G (fingers 1–3–5), first inversion E–G–C (1–2–5) and second inversion G–C–E (1–3–5).
- To find the root of an inverted chord, find the wider gap (5 keys). The note directly above it is the root.

- For smooth chord changes, keep shared notes still and move the other fingers to the nearest notes.
- The *bass* is the lowest part. The left hand plays the root low (alone, as a fifth or as an octave); the right hand plays the chord near middle C. Avoid full chords low down.
- *Accompaniment* patterns: block chords, broken chords, and the pop bounce.
- A slash chord such as C/E means a C chord with E in the bass. Slash chords let the bass move smoothly in a *stepwise bass line*.

Exercises

Exercise 4.1. Write out the three positions of F major (F, A, C) and of G major (G, B, D), starting from F4 and G4. Give the right-hand fingers for each.

Answer

F major: root position F4–A4–C5 (fingers 1–3–5); first inversion A4–C5–F5 (1–2–5); second inversion C5–F5–A5 (1–3–5).
G major: root position G4–B4–D5 (1–3–5); first inversion B4–D5–G5 (1–2–5); second inversion D5–G5–B5 (1–3–5).
The same shapes also exist an octave lower, and those are the ones you will use most: for example C4–F4–A4 for F and B3–D4–G4 for G.

Exercise 4.2. Use the wider-gap rule to name each chord and its position: (a) F♯4–A4–D5, (b) E♭4–A♭4–C5, (c) B♭3–E♭4–G4.

Answer

(a) F♯ to A is 3 keys; A up to D is 5 keys, the wider gap. The note above it is D. The chord is D major (D, F♯, A) in first inversion, because the third, F♯, is at the bottom.
(b) E♭ up to A♭ is 5 keys, the wider gap, so the root is A♭. The chord is A♭ major (A♭, C, E♭) in second inversion, with the fifth, E♭, at the bottom.
(c) B♭ up to E♭ is 5 keys, so the root is E♭. The chord is E♭ major (E♭, G, B♭) in second inversion.

Exercise 4.3. In the key of G, I–V–vi–IV is G, D, Em, C. Start on G4–B4–D5 and find a smooth right-hand version of the whole progression. At each change, say which notes stay.

Answer

G to D (D, F♯, A): the shared note is D5, so it stays. G4 moves down one key to F♯4, and B4 moves down two keys to A4. You hold F♯4–A4–D5.
D to Em (E, G, B): no note is shared. Move each finger to the nearest note of Em: F♯4 up to G4, A4 up to B4, D5 up to E5. You hold G4–B4–E5.
Em to C (C, E, G): G4 and E5 stay. B4 moves up one key to C5. You hold G4–C5–E5.
Then back to G: G4 stays, C5 moves down to B4, and E5 moves down to D5.

Exercise 4.4. A song sheet shows G, D/F♯, Em. What do your two hands play for D/F♯? What do the three left-hand notes do?

Answer

For D/F♯, the right hand plays a D major chord (D, F♯, A), for example F♯4–A4–D5, and the left hand plays F♯. The left-hand notes for the three chords are G, F♯, E. For example G2, F♯2, E2: the bass walks down one step at a time, like the stepwise bass line in this chapter.

Part III

Rhythm and Reading Music

Chapter 5

Rhythm: Counting Time

Hum the notes of *Happy Birthday* with long, random pauses between them, and nobody will recognize the tune. Now tap its rhythm on a table with one finger, with no notes at all. Most people will name the song within a few taps.

That is how much of music lives in time. So far this book has been about which keys to press. This chapter is about when to press them, and for how long.

You do not need to read music for this chapter. We will look at the symbols for long and short notes on their own, and count them out loud. Chapter 6 then puts them on the lines of written music.

5.1 The beat

Put on any song you like and tap your foot along with it. Your foot finds a steady pulse, like a clock ticking or a heart beating. That steady pulse is called the **beat**.

The beat keeps going at the same speed, even when the music pauses or a singer holds a long note. Here are eight steady beats played as clicks.

The speed of the beat is called the **tempo**. Musicians measure it in **beats per minute**, written BPM. At 60 BPM there is one beat every second, like a clock. At 120 BPM there are two beats every second, more like a brisk walk.

Listen to the same eight beats at 60 BPM and then at 120 BPM. The clicks are the same; only the speed changes.

A **metronome** is a small device or phone app that clicks a steady beat at any tempo you choose. It is the best rhythm teacher you can have, and many are free.

Try it at the keyboard

Set a metronome to 60 BPM, or play the first example above a few times. Tap your foot with the clicks. Then clap with them while you keep tapping. When that feels easy, try 90 BPM.

5.2 Rhythm: long and short sounds

The beat is steady. The notes on top of it are not: some are long and some are short. The pattern of long and short sounds is called the **rhythm**.

Sing the start of *Twinkle, Twinkle, Little Star* while you tap your foot. "Twin-kle twin-kle lit-tle" gets one tap for each part of a word. Then "star" lasts two taps. The beat never changed, but the last note was twice as long.

Here it is on the piano. Count "1, 2, 3, 4, 5, 6, 7–8" as you listen: the last note is held through two counts.

So beat and rhythm are two different things. The beat is the steady pulse you tap with your foot. The rhythm is the pattern your fingers play on top of it.

5.3 How long is each note?

In written music, the word note also means the small symbol that stands for one sound. Its shape tells you how long to hold the key down, counted in beats.

A note symbol is built from up to three parts. The **notehead** is the oval. The **stem** is the thin line going up or down from it. A **flag** is the little curl on the end of some stems. Figure 5.1 shows the five note lengths you will meet most often.



whole	half	quarter	eighth	sixteenth
4 beats	2 beats	1 beat	$\frac{1}{2}$ beat	$\frac{1}{4}$ beat

Figure 5.1: The five common note lengths. A hollow head means a long note; a filled head with more flags means a shorter one.

In most music, one quarter note lasts one beat. We will assume that for now. Each example below starts with four clicks to set the beat, called a **count-in**. The clicks then carry on under the notes, so you can hear how many beats each note lasts. Count along out loud.

- A **whole note** is a hollow oval with no stem. It lasts 4 beats.
- A **half note** is a hollow oval with a stem. It lasts 2 beats, so two of them take 4 beats.
- A **quarter note** is a filled oval with a stem. It lasts 1 beat: one note on every count.
- An **eighth note** has a filled oval, a stem and one flag. It lasts half a beat, so two fit in each beat.
- A **sixteenth note** has two flags. It lasts a quarter of a beat, so four fit in each beat.

Each step down the list halves the length. One whole note lasts as long as two half notes, or four quarter notes, or eight eighth notes. Figure 5.2 lines them up so you can see this.

When two or more eighth notes sit next to each other, their flags are usually joined into one thick line called a **beam**. Two beamed eighth notes look like a little bridge and together last one beat.

Sixteenth notes are joined by two beams.

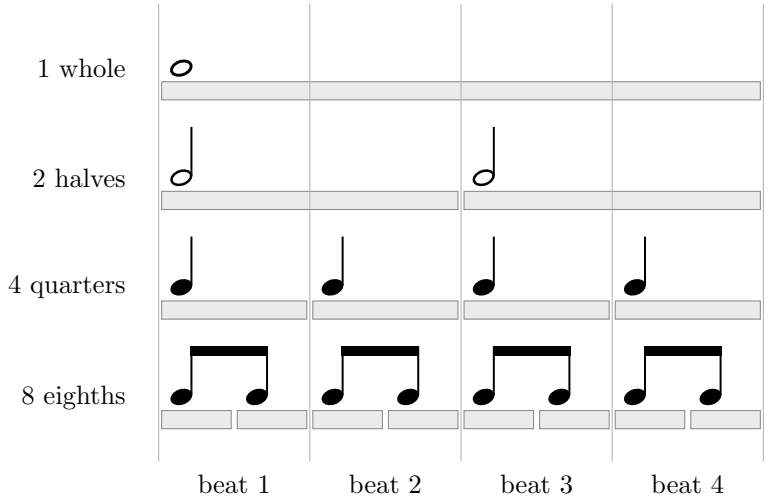


Figure 5.2: Each row fills the same four beats. Every step down splits each note into two shorter ones. Pairs of eighth notes are joined by a beam.

5.4 Rests: counting the silence

Music needs silence as well as sound. A **rest** is a symbol that means "play nothing here, and keep counting". Each note length has a matching rest, shown in Figure 5.3.



Figure 5.3: The four common rests. The whole rest hangs down from a line; the half rest sits on top of a line, like a hat.

- A **whole rest** is a small black block hanging below a line. It means a whole bar of silence (we meet bars in the next section). Here is a whole note, a whole rest, and another whole note.
- A **half rest** is the same block sitting on top of a line. It lasts 2 beats. A half note, a half rest, then the same again.
- A **quarter rest** looks like a squiggle standing up. It lasts 1 beat. Note, rest, note, rest.
- An **eighth rest** looks like a small 7 with a dot. It lasts half a beat.

To tell the whole and half rests apart, remember: the whole rest is heavier, so it hangs down; the half rest is lighter, so it sits up.

Common trap

Beginners often rush through rests, as if silence did not count. It does. Keep tapping your foot and counting out loud through every rest, so the next note comes in exactly on time.

5.5 Bars and time signatures

Long strings of beats are easier to follow in small groups. Written music splits the beats into equal boxes called **bars** (also called **measures**). A thin vertical line called a **bar line** marks the end of each bar. A double bar line, thin then thick, marks the end of the piece.

At the very start of the music you will see two numbers stacked on top of each other. This is the **time signature**. In this book we write it on one line, like "4/4" (say "four four").

- The top number says how many beats are in each bar.
- The bottom number says which note gets one beat. A 4 means the quarter note gets one beat; an 8 means the eighth note does.

Figure 5.4 shows four bars in 4/4 time. Each bar holds exactly four beats, filled in different ways.

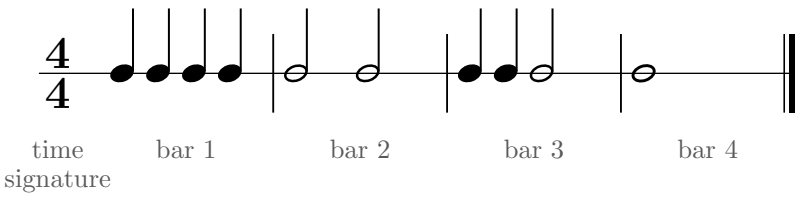


Figure 5.4: Four bars of 4/4 time. Every bar adds up to four beats: four quarters, two halves, two quarters and a half, or one whole note.

The first beat of every bar is a little stronger than the others, as if the music leans on it. That is how your ear feels where each bar begins.

4/4: four beats in a bar

Count "**1**, 2, 3, 4, **1**, 2, 3, 4" and lean a little on each 1. In the clicks below, the first beat of each bar is higher.

Most pop, rock and folk songs are in 4/4. It is so common that it is also called **common time**, and is sometimes written as a large letter C instead of two numbers.

3/4: three beats in a bar

Count "**1**, 2, 3, **1**, 2, 3". This is the swinging feel of a waltz.

Happy Birthday and *Amazing Grace* are both in 3/4. Hum one of them and count along: the strong "1" keeps coming back every three beats.

6/8: two groups of three

In 6/8 there are six eighth notes in each bar, grouped as three and three. Count "**1** 2 3 **4** 5 6". Your foot taps only twice per bar, on 1 and on 4, so it feels like a gentle rocking. *Row, Row, Row Your Boat* is usually written in 6/8.

Here is the difference you can hear. Below are quick notes grouped the 3/4 way, in pairs "**1** and **2** and **3** and", for two bars. And here are the same quick notes grouped the 6/8 way, in threes "**1** 2 3 **4** 5 6". Same number of notes, a different swing. Figure 5.5 shows both.

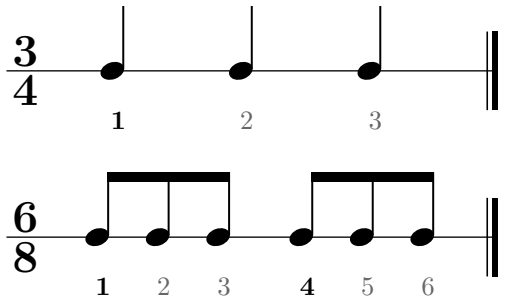


Figure 5.5: One bar of 3/4 (three beats) and one bar of 6/8 (six eighth notes beamed in two groups of three). The bold counts are where your foot taps.

5.6 Counting out loud

The most useful rhythm habit is simple: count out loud while you play. Say the beat numbers on the beats. Between them, say "and", exactly halfway to the next beat:

1 and 2 and 3 and 4 and

On paper this is written "1 & 2 & 3 & 4 &".

Quarter notes fall on the numbers. Eighth notes fall on the numbers and on every "and". Listen to four quarter notes and then eight eighth notes, and say "1 and 2 and 3 and 4 and" with both.

Now mix them. Try a quarter, two eighths, a quarter and a quarter. You play on "1, 2 and, 3, 4".

For sixteenth notes, split each beat into four and say "1 e and a" ("one ee and uh"). You will not need these much at first.

Try it at the keyboard

Clap each pattern while counting "1 and 2 and 3 and 4 and" out loud. Start very slowly.

1. Clap on every number: four quarter notes.
2. Clap on every number and every "and": eight eighth notes.
3. Clap on 1 and 3 only, holding your hands together until the next clap: two half notes.

5.7 Dotted notes

A small dot right after a note makes it longer. The rule: the dot adds half of the note's own length.

- A **dotted half note** lasts 2 + 1 = 3 beats. It fills a whole bar of 3/4.
- A **dotted quarter note** lasts 1 + a half = one and a half beats.

The dotted quarter is the one that confuses people, and counting out loud fixes it. It is almost always followed by an eighth note, and together they fill two beats. Hold the dotted quarter through "1 and 2", then play the eighth note on the "and" after 2. Figure 5.6 shows this.

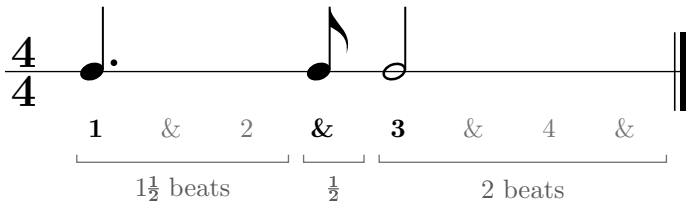


Figure 5.6: A dotted quarter, an eighth and a half note. The bold counts are the ones you play on: 1, the "and" after 2, and 3.

Listen to that bar: you hear notes on 1, on the "and" of 2, and on 3.

You know this rhythm already. In the chorus of *Jingle Bells*, "jin-gle all the way" uses it: "all" is a dotted quarter and "the" is an eighth.

5.8 Ties and slurs

A **tie** is a curved line joining two notes of the same pitch. Play the first note and hold it down for both lengths added together. Do not play the second one again.

Ties are how music holds a note across a bar line. In Figure 5.7, a half note is tied to a quarter note in the next bar, so you play once and hold for 2 + 1 = 3 beats. Listen: C, D, then E held for three beats, then D and C.

A curved line over notes of different pitches looks similar but means something else. It is a **slur**, and it asks you to play those notes smoothly, with no gap between them. Musicians call this smooth, connected playing **legato**.

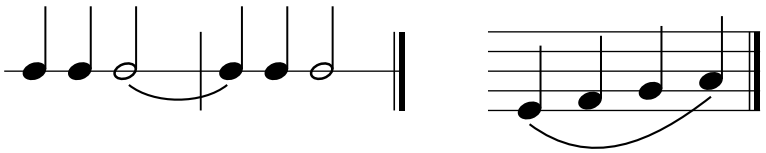


Figure 5.7: Left: a tie joins two notes of the same pitch across a bar line. Right: a slur arches over different notes.

Pause

Same pitch under the curve: it is a tie, so hold the note. Different pitches under the curve: it is a slur, so play each note, smoothly connected.

5.9 Syncopation

Most of the time, important notes land on the beat. Sometimes a note arrives on an "and" instead and the beat right after it is skipped. That surprise is called **syncopation**, and it gives music a bounce.

Count "1 and 2 and 3 and 4 and" and play on 1, on the first "and", on the next "and", then on 3 and 4. Beat 2 is never struck; the note before it is held through it.

You hear syncopation all the time in pop, jazz, funk and ragtime. When a rhythm feels tricky, write the counts under it and clap it slowly.

5.10 Two hands, one rhythm

Playing different rhythms with two hands sounds hard. The trick is to stop thinking of two separate rhythms and make them into one.

Write the counts "1 & 2 & 3 & 4 &" in a row. Under each count, mark what happens:

- T for together: both hands play.
- R for right hand only.
- L for left hand only.
- a dash (–) when nothing new is played.

Here is an example. The left hand plays two half notes, on 1 and 3. The right hand plays a quarter, two eighths and a half note, on 1, 2, the "and" of 2, and 3.

count	1	&	2	&	3	&	4	&
right hand	x		x	x	x			
left hand	x				x			
together	T	–	R	R	T	–	–	–

Now the two hands are one pattern: "T, –, R, R, T, –, –, –". Tap it on your knees while counting out loud, right hand on your right knee and left hand on your left. Then try it at the piano. In this recording the left hand plays C3 and the right hand plays E, D, E, C.

Try it at the keyboard

Tap the pattern above on your knees ten times, slowly, counting out loud. Then speed up a little. If it falls apart, slow down again: slow and steady always beats fast and shaky.

What to remember

- The *beat* is the steady pulse you tap your foot to. The *tempo* is its speed, in beats per minute (BPM).
- The *rhythm* is the pattern of long and short notes played over the beat.
- Note lengths: whole = 4 beats, half = 2, quarter = 1, eighth = half a beat, sixteenth = a quarter of a beat.
- A *rest* is a measured silence. Each note length has a matching rest. Keep counting through rests.
- *Bars* split the music into equal groups of beats. The *time signature* (4/4, 3/4, 6/8) tells you how many beats are in each bar.
- Count out loud: "1 and 2 and 3 and 4 and". A dot adds half a note's length; a *tie* joins two notes into one long one.
- *Syncopation* puts notes on the "and" and skips the beat. For two hands, combine them into one T/R/L pattern.

Exercises

Exercise 5.1. How many quarter notes last as long as one whole note? How many eighth notes last as long as one half note?

Answer

A whole note lasts 4 beats and a quarter note lasts 1, so four quarter notes fill one whole note. A half note lasts 2 beats and each eighth note lasts half a beat, so four eighth notes fill one half note. You can check both in Figure 5.2.

Exercise 5.2. A bar of 4/4 starts with a half note and a quarter note. What single note would complete the bar? What single rest would complete it?

Answer

The half note (2 beats) and the quarter note (1 beat) use 3 of the 4 beats. One beat is left, so a quarter note completes the bar. A quarter rest would also complete it, with one beat of silence at the end.

Exercise 5.3. In 4/4, a bar has a dotted quarter note, an eighth note and a half note. Write "1 & 2 & 3 & 4 &" and circle the counts where you play.

Answer

You play on 1, on the "&" after 2, and on 3. The dotted quarter is held through "1 & 2" (one and a half beats). The eighth note takes the "&" of 2. The half note starts on 3 and is held through "3 & 4 &".

Exercise 5.4. Six quick notes can be grouped as 3/4 or as 6/8. In your own words, how does each one feel when you tap your foot?

Answer

In 3/4 the six notes come in three pairs: “1 and 2 and 3 and”, and your foot taps three times per bar. In 6/8 they come in two groups of three: “1 2 3 4 5 6”, and your foot taps only twice, on 1 and 4. The 3/4 version feels like a waltz; the 6/8 version feels like rocking in a boat.

Exercise 5.5. In one bar of 4/4, the left hand plays half notes on 1 and 3. The right hand plays a quarter note, two eighth notes, a dotted quarter note and an eighth note. Fill in the T/R/L/– row for “1 & 2 & 3 & 4 &”.

Answer

First find where the right hand plays. The quarter note is on 1. The two eighths are on 2 and the “&” of 2. The dotted quarter starts on 3 and lasts one and a half beats, through “3 & 4”. The last eighth note falls on the “&” of 4. The left hand plays on 1 and 3.
So the combined row is: T, –, R, R, T, –, –, R. Tap it on your knees, then listen.

Chapter 6

Reading Notes on the Staff

In Chapter 5 you learned how long each note lasts. Written music also shows which key to play. It does this with a simple ladder of lines, where higher on the page means higher on the piano.

This chapter teaches you to read that ladder calmly. You will not memorize every note. Instead you will learn a few landmark notes and read the rest by counting small steps from them, the same way you find a house from the nearest street corner.

6.1 The staff: a ladder of lines and spaces

Music is written on five horizontal lines called a **staff**. The five lines leave four spaces between them. We count both from the bottom: line 1 is the bottom line, and space 1 is the space directly above it.

A notehead can sit *on a line*, with the line running through its middle. Or it can sit *in a space*, between two lines. Figure 6.1 shows notes climbing the staff: line, space, line, space, and so on.

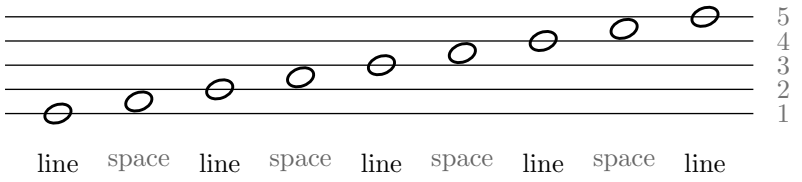


Figure 6.1: A staff has five lines and four spaces. Each note here is one step higher than the one before: line, space, line, space.

Here is the key idea of this chapter. Each line and each space stands for one white key. Moving up from a line to the very next space (or from a space to the next line) moves up one white key, to the next letter of the musical alphabet. We call that move one **step**.

So climbing the staff is the same as walking up the white keys: C, D, E, F, G, A, B, C, and so on. Higher on the staff means further right on the keyboard, and higher in sound. A step on the staff counts letters, not keys: it can be a half step on the keyboard (E to F, B to C) or a whole step (C to D).

The black keys do not get their own lines or spaces. They are written with sharp and flat signs,

which we meet later in this chapter.

6.2 Clefs: which note is which

A bare staff does not say which letter each line is. That is the job of a **clef**, a symbol drawn at the start of every staff. A clef marks one line as a particular note. Everything else follows by counting steps.

The **treble clef** is the tall, curly sign. Its inner curl wraps around line 2, and marks that line as G4, the first G above middle C. That is why it is also called the G clef. The treble staff holds higher notes, usually played by the right hand.

The **bass clef** looks like a backwards C with two dots. The two dots sit on either side of line 4, and mark that line as F3, the first F below middle C. That is why it is also called the F clef. The bass staff holds lower notes, usually played by the left hand.

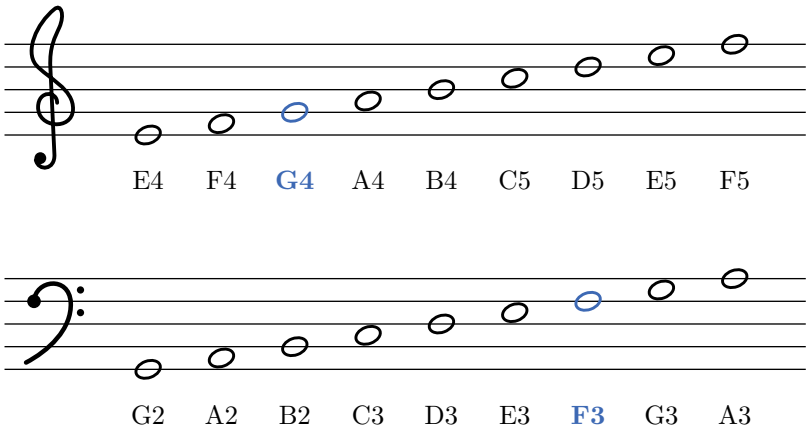


Figure 6.2: The treble clef marks G4 (line 2) and the bass clef marks F3 (line 4). Every other line and space follows by stepping up or down.

From those two notes you can fill in the rest by stepping, as in Figure 6.2.

- Treble staff, lines from the bottom: E4, G4, B4, D5, F5. Spaces: F4, A4, C5, E5.
- Bass staff, lines from the bottom: G2, B2, D3, F3, A3. Spaces: A2, C3, E3, G3.

You may have heard rhymes for these, like “Every Good Boy Does Fine” for the treble lines and “FACE” for the treble spaces. They work, but reciting a rhyme for every note is slow. The landmark method below is quicker once it clicks, and you can always fall back on a rhyme.

6.3 The grand staff and middle C

Piano music uses two staves at once, joined at the left by a long line or a curly bracket. Together they are called the **grand staff**: treble on top for the right hand, bass below for the left hand.

The two staves fit together like one tall ladder. The top line of the bass staff is A3. One step up is B3, in the space directly above it. One more step up is middle C, C4. One more step is D4, in the

space directly below the treble staff. And one more step lands on E4, the bottom line of the treble staff.

Middle C sits between the two staves, so it gets its own short line drawn through it. A short line like this, added above or below a staff, is called a **ledger line**. Figure 6.3 shows middle C on its ledger line, halfway between the staves.

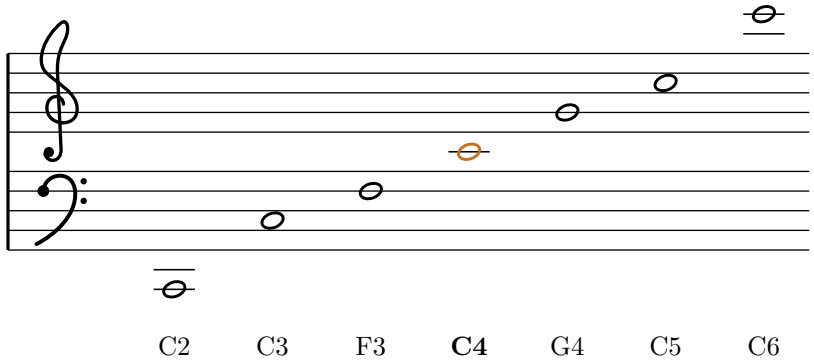


Figure 6.3: The grand staff, drawn as one ladder. Middle C (C4) sits on its own ledger line between the staves. The other notes shown are the landmarks of the next section.

In printed music the two staves are drawn further apart, to leave room. Then middle C appears in one of two places: on a ledger line directly below the treble staff when the right hand plays it, or directly above the bass staff when the left hand plays it. Both are the same key, as Figure 6.4 shows.

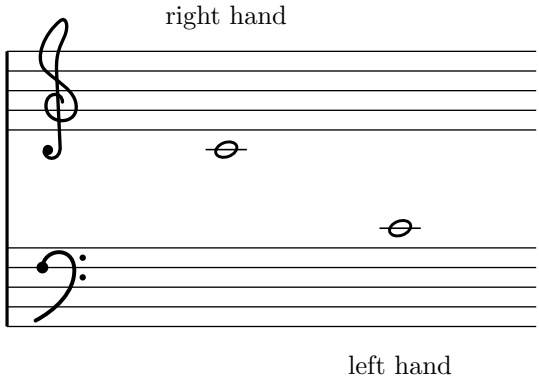


Figure 6.4: In printed music, middle C is written directly below the treble staff (for the right hand) or directly above the bass staff (for the left hand). Both mean the same key.

6.4 Ledger lines

Ledger lines work the same way above and below any staff. They extend the ladder: each extra short line is one more line of the staff, and the spaces between them count too.

Above the treble staff: the top line is F5. The space above it is G5, then A5 sits on the first ledger line, B5 above it, and C6 on the second ledger line. Below the treble staff: D4 hangs directly under the bottom line, then C4 on one ledger line, then B3 and A3.

The bass staff works the same way. Below its bottom line (G2) come F2, then E2 on the first ledger line, D2, and C2 on the second ledger line.

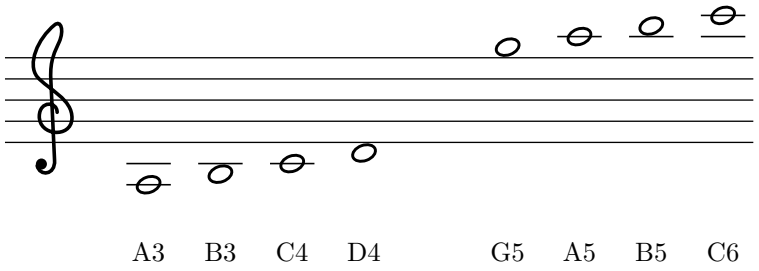


Figure 6.5: Ledger lines extend the treble staff downward (A3 to D4) and upward (G5 to C6). Count them like ordinary lines and spaces.

6.5 Landmark notes

Reading music fluently does not mean naming every note from scratch. It means knowing a handful of landmark notes by sight, and counting a step or two from the nearest one.

Here are five landmarks, all around middle C (they are in Figure 6.3):

- Middle C (C4), on its own ledger line between the staves.
- G4, the treble clef line (line 2 of the treble staff).
- F3, the bass clef line (line 4 of the bass staff).
- C5, in space 3 of the treble staff.
- C3, in space 2 of the bass staff.

Notice the mirror pattern. G4 is the second line up from the bottom of the treble staff; F3 is the second line down from the top of the bass staff. Both sit four steps away from middle C, one above and one below. C5 and C3 sit seven steps away, again one above and one below.

Hear the five landmarks from low to high.

From C3 up to C5, every white key is at most two steps from one of these five landmarks. To read a note, find the nearest landmark and count. One step goes from a line to the next space (or a space to the next line). Two steps go from a line to the next line, or from a space to the next space.

For notes further out, add four more landmarks: the bottom line of the bass staff (G2), the top line of the treble staff (F5), and the two Cs on second ledger lines, C2 below the bass staff and C6 above the treble staff. With all nine, every white key from C2 up to C6 is within two steps of a landmark.

A few examples:

- The next treble line above G4 is two steps up (G, A, B), so it is B4.
- The space directly above middle C is one step up, so it is D4.
- The next bass line below F3 is two steps down (F, E, D), so it is D3.

Try it at the keyboard

Put the thumb of your right hand on middle C and look at Figure 6.3. Play C4, then G4 with your little finger. Then put your left little finger on C3 and play F3 with your index finger (finger 2), then C3 again. Say each landmark's name out loud as you play it.

6.6 Sharps, flats and naturals on paper

Black keys are written with signs placed directly before the note, on the same line or space.

- A **sharp** sign (♯) means: play the key one half step higher. A note on the F4 space with a sharp is F♯4, the black key directly right of F4.
- A **flat** sign (♭) means: play the key one half step lower. A note on the B4 line with a flat is B♭4, the black key directly left of B4.
- A **natural** sign (♮) cancels a sharp or flat and brings the note back to the white key.

A sharp, flat or natural written in front of a note is called an **accidental**. It lasts until the end of the bar, for every note on that same line or space. At the next bar line it is forgotten.

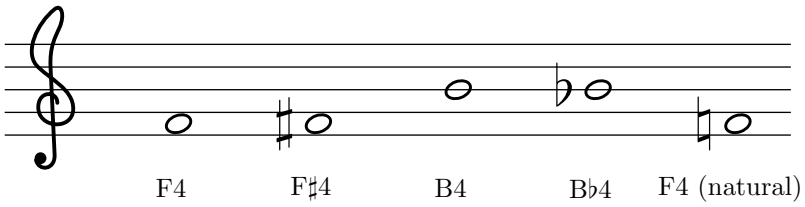


Figure 6.6: A sharp raises a note by a half step, a flat lowers it by a half step, and a natural cancels either one.

6.7 Key signatures

Chapter 2 showed that the scale of G major uses F♯ instead of F. A piece in the key of G (a piece built on that scale) would need a sharp in front of every F. Writing them all out would be tiring.

Instead, the sharps or flats a piece needs are written once, at the start of every staff, directly after the clef. This group is called the **key signature**. A sharp on the F line in the key signature means: every F in the piece, in every octave, is played as F♯, unless a natural sign says otherwise.

Figure 6.7 shows four key signatures. The sharps always appear in the same order, F C G D A E B, and the flats in the order B E A D G C F, as in Chapter 2.

Two quick tricks tell you the key:

- *Sharps:* take the last sharp and go up one half step. One sharp (F♯): up a half step is G, so G major. Two sharps (F♯, C♯): the last is C♯, up a half step is D, so D major.
- *Flats:* the second-to-last flat is the key. Two flats (B♭, E♭): the second-to-last is B♭, so B♭ major. One flat has no second-to-last, so learn that one by heart: one flat (B♭) means F major.

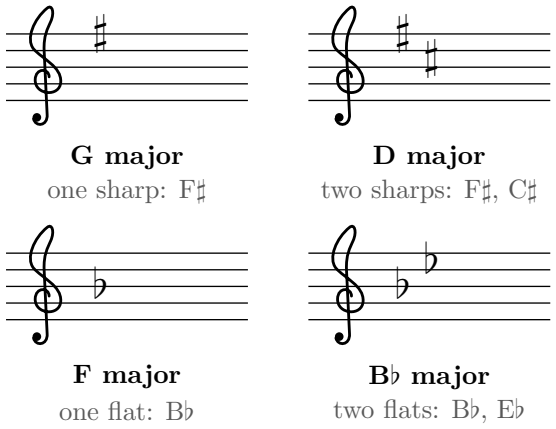


Figure 6.7: Four key signatures on the treble staff. The signs sit on the lines and spaces of the notes they change.

No sharps or flats at all means C major.

Each key signature is also shared by one minor key, its relative minor from Chapter 2. To find it, count down three half steps from the major key's name. G major and E minor share one sharp; C major and A minor share no sharps or flats. The last note of a piece usually tells you which one it is: a piece in E minor tends to end on E.

6.8 Intervals by shape

In Chapter 1 you measured an *interval*, the distance between two notes, in half steps. On the staff it is named by counting letters, including both ends. C up to E is a third, because it covers three letters: C, D, E.

The staff shows intervals as shapes, so you can read them without naming the notes. Figure 6.8 shows the common ones, all starting from E4 on the bottom line.

- A step (a second) goes from a line to the very next space, or a space to the next line.
- A **skip** (a third) goes from a line to the next line, or a space to the next space.
- A fourth goes from a line to the second space above it.
- A fifth goes from a line to a line two away, skipping one line in between.
- An octave (an eighth) goes from E to the next E, from a line to a space.

There is a handy pattern. Odd intervals (third, fifth, seventh) go from a line to a line, or a space to a space. Even intervals (second, fourth, sixth, octave) switch between line and space.

This links directly to your fingers. With one finger on each of five neighboring white keys, a step moves to the next finger, a skip skips one finger, and a fifth goes from thumb to little finger. Once your hand is in place, you can read a melody by its ups and downs alone.

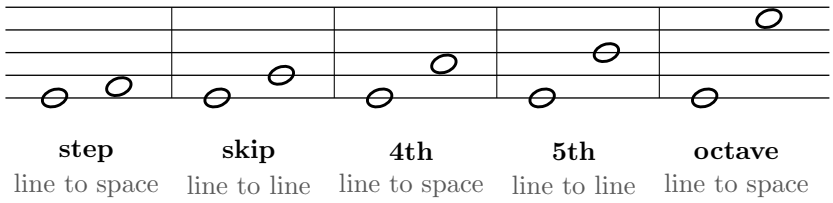


Figure 6.8: Intervals as shapes: a step, a skip, a fourth, a fifth and an octave, each starting from E4 on the bottom line.

6.9 Chord shapes on the staff

A chord written on the staff is a stack of noteheads, one above the other, to be played together. The three positions of a chord from Chapter 4 each have their own shape, as Figure 6.9 shows.

- *Root position* (C–E–G) stacks on three lines in a row, or three spaces in a row. It looks like a neat snowman.
- *First inversion* (E–G–C) has two notes close together at the bottom and a wider gap on top.
- *Second inversion* (G–C–E) has a wider gap at the bottom and two notes close together on top.

The gap tells you where the root is. In both inversions, the root (the note the chord is named after) is the note directly above the gap.

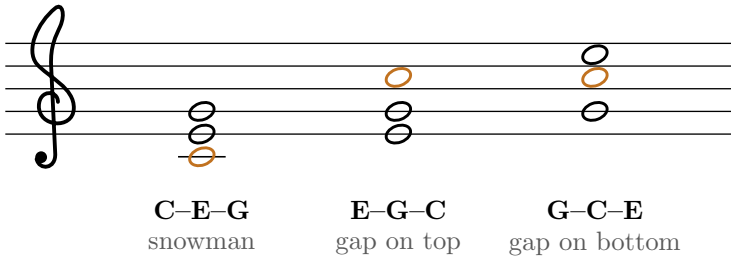


Figure 6.9: The C major chord in its three positions. The root C is colored; in each inversion it sits directly above the gap.

6.10 Calm habits for reading new music

Reading a piece you have never seen is called **sight-reading**. Here are five habits that make it calmer.

1. *Look before you play.* Spend half a minute on the page first. Check the key signature (which black keys are in use) and the time signature (how to count). Find the first note for each hand and put your hands in place.
2. *Pick a slow tempo and keep the beat.* Choose a speed at which the hardest bar feels easy. Count one bar out loud before you start.
3. *Look a little ahead.* While you play one note, your eyes should already be on the next. Try not to look down at your hands; feel for the black keys instead.

4. *Do not stop for wrong notes.* A missed note is forgotten in a moment; stopping breaks the music. Keep the beat and land on the next note.
5. *Practice slowly.* It is better to play four bars slowly and steadily than fast with stops. Speed comes by itself once the notes feel familiar.

What to remember

- A *staff* has five lines and four spaces. Each line or space is one white key; moving up one *step* (line to next space) moves up one letter.
- The *treble clef* marks G4 on line 2; the *bass clef* marks F3 on line 4. Together they make the *grand staff*.
- Middle C sits on a short *ledger line* between the staves. Ledger lines extend any staff up or down.
- Landmarks: C3, F3, C4, G4 and C5 (plus G2, F5, C2 and C6 further out). Read other notes by counting steps from the nearest one.
- A *sharp* raises a note a half step, a *flat* lowers it, a *natural* cancels either.
- The *key signature* lists the sharps or flats used all through a piece. Sharps: last sharp up a half step. Flats: second-to-last flat.
- *Intervals* have shapes: step = line to space, skip = line to line. Chords show as a snowman, gap on top, or gap on bottom; the root sits directly above the gap.

Exercises

Exercise 6.1. Use the landmarks. Name the note (a) on the next treble line above G4, (b) in the space directly below middle C, (c) on the next bass line below F3.

Answer

(a) From G4, the next line up is two steps higher: G, A, B. It is B4.
(b) From middle C, one step down is B3, which sits in the space directly under middle C's ledger line.
(c) From F3, the next line down is two steps lower: F, E, D. It is D3.

Exercise 6.2. Name the note on the first ledger line above the treble staff, and the note on the second ledger line below the bass staff.

Answer

The top line of the treble staff is F5. One step up is G5 (in the space above), and one more step is the first ledger line: A5.
The bottom line of the bass staff is G2. Stepping down: F2 (space), E2 (first ledger line), D2 (space), C2 (second ledger line). It is C2, one of the far landmarks.

Exercise 6.3. A key signature has three sharps. Which major key is it? Another has three flats. Which major key is that?

Answer

Sharps always come in the order F, C, G, so three sharps are F♯, C♯ and G♯. The last one is G♯; one half step up is A. The key is A major.
Flats come in the order B, E, A, so three flats are B♭, E♭ and A♭. The second-to-last is E♭, so the key is E♭ major.

Exercise 6.4. On the treble staff, one note sits on the bottom line and another on the middle line. What is the interval between them? Which fingers would play it in a five-finger position starting on the lower note?

Answer

Line to line, with one line skipped in between, is a fifth. The notes are E4 and B4: E, F, G, A, B is five letters. With your thumb on E4, the fifth is under your little finger: fingers 1 and 5.

Exercise 6.5. Three notes are stacked on the treble staff: E4 (line 1), A4 (space 2) and C5 (space 3). Which shape is this: snowman, gap on top, or gap on bottom? What is the chord?

Answer

From E4 to A4 is a fourth, a wider gap; from A4 to C5 is a skip. So the gap is on the bottom, and this is a second inversion. The root is the note directly above the gap: A. The notes A, C and E make an A minor chord, here played as E–A–C, with fingers 1, 3 and 5.

Part IV

Playing Real Music

Chapter 7

Your First Pieces

You now know where the notes are, how scales and chords are built, how to count rhythm and how to read the staff. This chapter puts all of it together in real music.

We will learn three pieces, step by step. First, the famous melody of Beethoven's *Ode to Joy*, then the same tune with a left hand. Next, the opening of *Amazing Grace*, a slow song in 3/4 time. Last, a four-chord accompaniment that fits hundreds of pop songs. Both melodies are old enough to be free for anyone to play.

Each piece is written three ways: as letters with finger numbers, on the staff, and as sound. Use whichever helps most, and let the others check your work.

7.1 Getting ready: the C position

Most first pieces use one hand position, so your fingers never have to search. In the **C position**, each finger of the right hand rests on one white key, starting from middle C: thumb (finger 1) on C4, then D4, E4, F4, and little finger (finger 5) on G4.

The left hand mirrors this, an octave lower. Little finger (5) on C3, then 4 on D3, 3 on E3, 2 on F3, and thumb (1) on G3. Figure 7.1 shows both hands.

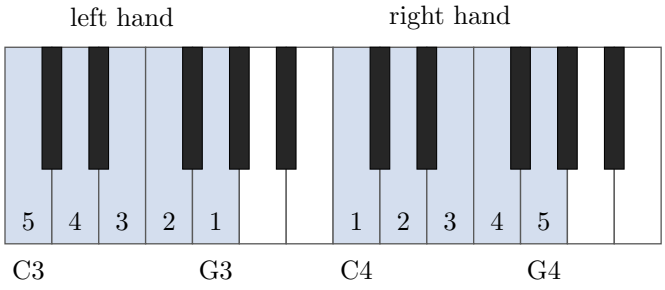


Figure 7.1: The C position. The numbers are the fingers that rest on each key: thumbs are 1, little fingers are 5.

Sit so your forearms are about level with the keys. Curve your fingers gently, as if holding a ball, and play with the fleshy tips. Keep your shoulders loose.

Try it at the keyboard

Place your right hand in C position. Play C, D, E, F, G and back down, saying the finger numbers out loud: 1, 2, 3, 4, 5, 4, 3, 2, 1. Do the same with the left hand: 5, 4, 3, 2, 1, 2, 3, 4, 5.

7.2 Ode to Joy: the right hand

Beethoven wrote this tune for the last movement of his Ninth Symphony, in 1824. It moves mostly by steps, so it sits comfortably under a hand in C position. Its time signature is 4/4, and almost every note is a quarter note: one note per beat.

The tune has two phrases. A **phrase** is a musical sentence, a few bars long, that ends with a feeling of a pause. Here is the first one, bar by bar:

	bar 1	bar 2	bar 3	bar 4
notes	E E F G	G F E D	C C D E	E. D D
fingers	3 3 4 5	5 4 3 2	1 1 2 3	3 2 2

Bars 1 to 3 are four quarter notes each. Bar 4 is the rhythm from Chapter 5: a dotted quarter (the "E."), an eighth note, and a half note. Count "1 and 2 and 3 and 4 and": E on 1, the first D on the "and" after 2, and the last D on 3, held to the end of the bar.

Listen to the first phrase. It ends on D, which sounds a little unfinished, like a question.

The second phrase starts the same way. Only the last bar changes, and it ends on C, which sounds like an answer:

	bar 5	bar 6	bar 7	bar 8
notes	E E F G	G F E D	C C D E	D. C C
fingers	3 3 4 5	5 4 3 2	1 1 2 3	2 1 1

Listen to the second phrase, then to both phrases together.

Figure 7.2 shows the same melody on the treble staff, with finger numbers above the notes and letters below. Notice how the notes climb and fall by steps, line to space to line.

Try it at the keyboard

Play the first phrase very slowly, at about half the speed of the recording, counting out loud. Do not worry about mistakes; keep going to the end. Then play it three more times. When it feels steady, learn the second phrase, which only changes in the last bar.

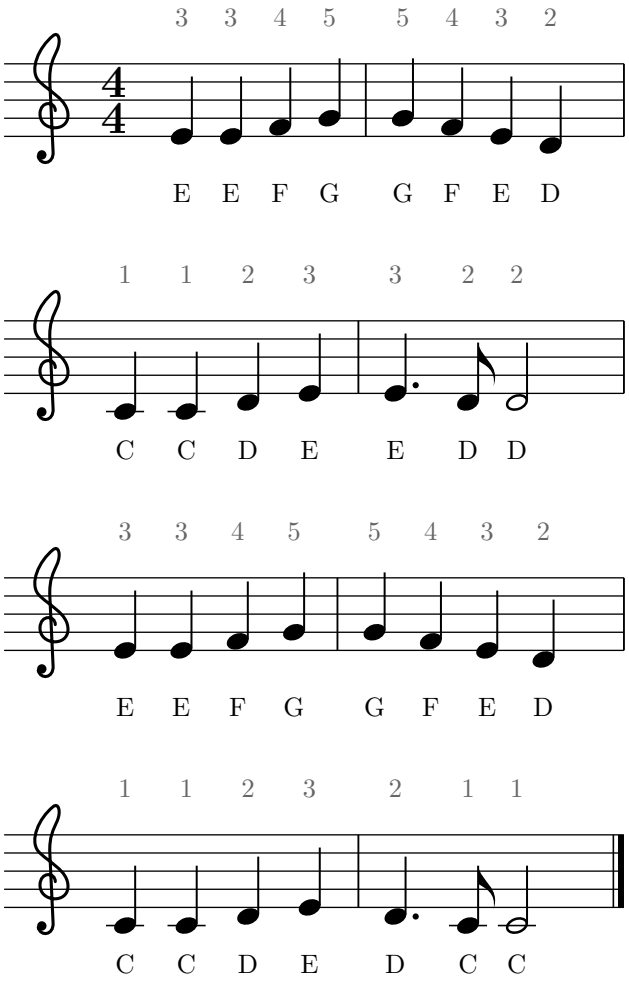


Figure 7.2: Ode to Joy, both phrases, in C position. Finger numbers are above the staff and note names below.

7.3 Ode to Joy: adding the left hand

Now give the melody a floor to stand on. We add the left hand in two stages: first single bass notes, then full chords.

Stage 1: bass notes

Put your left hand in C position. You need only two keys: C3 under your little finger (5) and G3 under your thumb (1). C goes with the C major chord and G with the G major chord, the I and V chords of the key of C from Chapter 3.

Play one bass note at the start of each bar, together with the first melody note, and hold it while the right hand carries on. In the last bar of each phrase, the bass changes halfway, on beat 3:

bar	1	2	3	4	5	6	7	8
left hand	C	G	C	C, then G	C	G	C	G, then C

Listen to the first phrase with bass notes and the second.

Practice each hand alone first. Then put them together very slowly, using the T/R/L idea from Chapter 5: on the first beat of each bar both hands play (T), and on the other beats only the right hand plays (R). The one exception is beat 3 of each phrase's last bar, where the bass changes: that beat is T as well.

Stage 2: chords

When the bass notes feel easy, replace them with three-note chords. The C chord is C3–E3–G3, with fingers 5, 3 and 1.

For the G chord, use the smooth change from Chapter 4. Keep your thumb on G3. Move your middle finger down one white key, from E3 to D3, and your little finger down a half step, from C3 to B2. That gives B2–D3–G3, a G chord in first inversion, with fingers 5, 3, 1. Your hand hardly moves: C chord, G chord, C chord.

Play each chord where the table above has its bass note. Here is the first phrase with chords and the second.

7.4 Amazing Grace: a song in 3/4

Amazing Grace is a hymn from the 1770s, sung to an American folk tune. It is in 3/4 time, so count "**1**, 2, 3" with a gentle lean on each 1. We will learn its first line, "Amazing grace, how sweet the sound, that saved a wretch like me".

The pickup note

The song does not start on beat 1. Its first word, “A-”, comes on beat 3, directly before the first full bar. A note like this, before the first bar line, is called a **pickup**. Count “1, 2” silently, play the pickup on “3”, and the next note lands on the strong beat 1.

The right hand

This song is in the key of G, which has one sharp, F♯. You can see it as the key signature after the clef in Figure 7.3. The first line happens not to use F at all, so you will not need a black key yet.

Place your right thumb on D4, finger 2 on E4, finger 3 on G4, finger 4 on A4 and finger 5 on B4. (Your hand skips over F♯.) Only the very last note, D5, is out of reach: lift your hand and play it with finger 5. A small move like this is normal.

bar	words	notes	fingers	beats	left hand
pickup	A-	D	1	1	–
1	-maz-ing	G, B G	3, 5 3	2, half + half	G
2	grace, how	B, A	5, 4	2, 1	G
3	sweet the	G, E	3, 2	2, 1	C
4	sound, that	D, D	1, 1	2, 1	G
5	saved a	G, B G	3, 5 3	2, half + half	G
6	wretch like	B, A	5, 4	2, 1	G
7	me	D (high)	5	3	D

“-maz-ing” has a half note on “maz”, then two eighth notes, B and G, on “-ing”. The last note is a dotted half note, held for the whole bar. Listen.

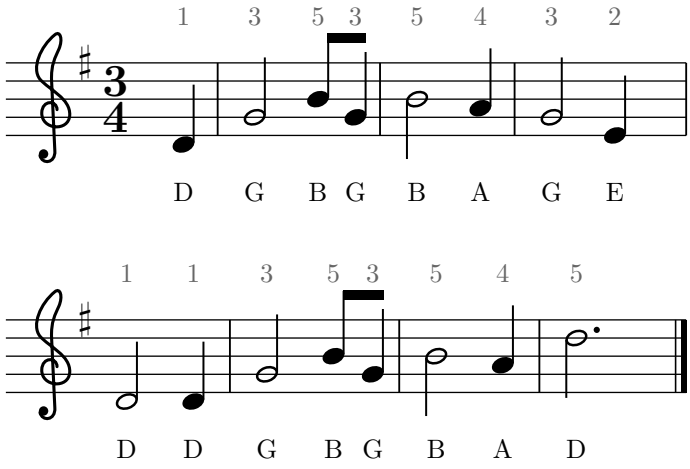


Figure 7.3: The first line of Amazing Grace in 3/4, starting with a one-beat pickup. The sharp after the clef is the key signature of G major. Finger numbers are above the staff and note names below.

The left hand

The left hand plays one chord at the start of each full bar and holds it for all three beats. You need three chords, shaped so your hand barely moves:

- G chord: G2–B2–D3, fingers 5, 3, 1.
- C chord: G2–C3–E3, fingers 5, 2, 1. Keep G2, and move the other two fingers up one white key each.
- D chord: F♯2–A2–D3, fingers 5, 3, 1. Keep D3, and move the other two down: G2 down one key to F♯2, and B2 down one white key to A2.

Here they are in order: G, C, G, D.

The "left hand" column of the table shows which chord goes in each bar. The pickup has no chord. Listen to both hands together.

7.5 Four chords for a hundred songs

Many pop songs, old and new, use the same four chords over and over: I, V, vi and IV. In the key of C these are C, G, A minor (written Am) and F. Once you can loop them smoothly, you can accompany a huge number of songs, and make up your own.

The right hand: smooth shapes

Play every chord with fingers 1, 3 and 5, and choose the positions from Chapter 4 so your hand hardly moves, as in Figure 7.4:

- C: G3–C4–E4 (second inversion).
- G: G3–B3–D4. Keep your thumb on G; the other two fingers step down.
- Am: A3–C4–E4. All three fingers step up one white key.
- F: A3–C4–F4 (first inversion). Keep A and C; only the little finger moves, up from E to F. Fingers 1, 3 and 5 keep C under the same finger here; the 1–2–5 fingering from Chapter 4 works too.

Then back to C: the thumb steps down to G, the little finger steps down to E, and C stays.

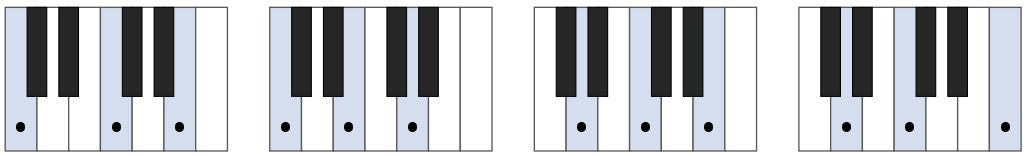


Figure 7.4: The four chords C, G, Am and F under the right hand, each played with fingers 1, 3 and 5. Neighboring chords share keys, so the hand barely moves.

The left hand: roots

The left hand plays the root of each chord, low down: C3 for C, G2 for G, A2 for Am, and F2 for F.

Two patterns

Pattern 1 is the simplest. Play the left-hand root and the right-hand chord together on beat 1, and hold them for the whole bar. One chord per bar, four bars, then end on C.

Pattern 2 adds a pulse. The left hand plays its root on beat 1 and holds it. The right hand plays the chord on every beat: 1, 2, 3, 4.

Try it at the keyboard

Loop pattern 1 four times without stopping, counting "1, 2, 3, 4" in every bar. Then try pattern 2. Once it is steady, hum any melody you like over it, or improvise one with your right hand on the white keys while your left hand plays the roots.

7.6 A simple practice routine

Short, regular practice works far better than long, rare sessions. Fifteen to twenty minutes a day is plenty to make steady progress. Here is one way to spend them:

minutes	what	how
2	warm-up	C position, up and down, each hand, slowly
3	scales	C major and G major, hands separately (Chapter 2)
3	chords	the four-chord loop, pattern 1 then pattern 2
2	rhythm	clap and count one pattern from Chapter 5
3	reading	a few bars of new music, slowly, without stopping
5	your piece	one phrase at a time: hands alone, then together

A few habits make every minute count:

- Practice slowly enough to play it right. Speed comes from many correct, calm repetitions.
- Work on the hard bar, not the easy ones. Start your practice there, not always from the beginning.
- Use a metronome now and then, so your sense of the beat stays honest.
- Finish each session by playing something you enjoy, all the way through.

7.7 Where to go next

You have the tools to keep learning on your own. Here are some good next steps.

- *More pieces.* Many lovely beginner pieces are free to play: the *Minuet in G* from the notebook Bach kept for his wife Anna Magdalena, the opening of Beethoven's *Für Elise*, and folk songs and hymns of every kind. Look for "easy piano" arrangements.
- *Lead sheets.* A lead sheet gives only the melody and chord symbols. Play the melody with your right hand and the chords with your left, as you did with Ode to Joy. This is how many pianists play popular songs.
- *More keys.* Use the scale recipe from Chapter 2 to play Ode to Joy and the four-chord loop in G, F and D.
- *A teacher.* Even a few lessons help a lot with posture and hand shape, which are hard to judge on your own.
- *Listen and record.* Record yourself on your phone now and then, and listen back. You will hear things you miss while playing, and you will hear your progress.

What to remember

- In the *C position*, the right thumb rests on middle C and the fingers cover C to G; the left hand mirrors it from C3 to G3.
- A *phrase* is a musical sentence. Ode to Joy has two phrases that differ only in their last bar.
- Add the left hand in stages: single bass notes first, then chords in smooth positions.
- A *pickup* is a note before the first full bar. In Amazing Grace it falls on beat 3.
- The chords I, V, vi and IV (C, G, Am, F in C) fit countless songs. Smooth positions keep the hand still.
- Practice a little every day, slowly, starting with the hardest part.

Exercises

Exercise 7.1. Move Ode to Joy to G position: right thumb on G4, then A4, B4, C5 and little finger on D5. Write the letters of the first phrase. Do the finger numbers change?

Answer

Every note moves up by the same amount, from C position to G position: C becomes G, D becomes A, E becomes B, F becomes C, and G becomes D. So the first phrase becomes:
B B C D, D C B A, G G A B, B. A A.
The finger numbers stay exactly the same (3 3 4 5, 5 4 3 2, 1 1 2 3, 3 2 2), because each finger still plays the same step of the hand position. Listen.

Exercise 7.2. In the last bar of the first phrase of Ode to Joy (E. D D), how many beats does each note get? Check that they add up to a full bar of 4/4.

Answer

The dotted quarter E gets one and a half beats, the eighth note D gets half a beat, and the half note D gets 2 beats. One and a half plus a half is 2, and 2 more makes 4: a full bar.

Exercise 7.3. Amazing Grace starts with a pickup. On which count is the pickup played, and why does the song not start on 1?

Answer

The pickup "A-" is played on count 3. In the words "a-MAZ-ing grace", the stress falls on "maz", not on "a". The strong first beat of the bar belongs on the stressed syllable, so the weak "a" comes directly before it, on the last beat of an incomplete bar.

Exercise 7.4. In the key of G, the I, V, vi and IV chords are G, D, Em and C. Find right-hand positions where neighboring chords share as many keys as possible, starting from G as D4–G4–B4.

Answer

One smooth set is:

- G: D4–G4–B4.
- D: D4–F♯4–A4. Keep D; the other two fingers step down.
- Em: E4–G4–B4. All three fingers step up.
- C: E4–G4–C5. Keep E and G; only the little finger moves, from B up to C.

Then back to G: E steps down to D, G stays, and C steps down to B. The whole loop.

Exercise 7.5. You have only ten minutes today. Which parts of the routine would you keep?

Answer

There is no single right answer. A good plan keeps a short warm-up (1 minute), spends most of the time on your piece (6 minutes, starting with the hardest bar), and ends with a few minutes of chords or reading. What matters most is playing a little every day, slowly and without tension.